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For correspondence:

tktang@ibms.sinica.edu.tw

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The Multifaceted Role of EXOC6A in Ciliogenesis

Te-Lin Lin^{1,2}, Chien-Ting Wu^{2,3}, Tang K Tang^{1,2} 

¹Taiwan International Graduate Program in Molecular Medicine, National Yang Ming Chiao Tung University and Academia Sinica, Taipei, Taiwan • ²Institute of Biomedical Sciences, Academia Sinica, Taipei, Taiwan • ³Department of Microbiology, University of Texas Southwestern Medical Center, Dallas, United States

eLife Assessment

This **important** study elucidates the role of the exocyst component EXOC6A at distinct stages of ciliogenesis, which advances our understanding of ciliary membrane remodeling and cilium formation. The authors provide **compelling** evidence through high quality light and electron microscopic imaging, and careful analysis of knockout cell lines, that EXOC6A interacts with myosin-Va and is dynamically recruited via dynein-, microtubule-, and actin-dependent mechanisms, to support proper formation of the ciliary membrane. The study will be of interest to cell biologists and other researchers interested in vesicular trafficking, organellar membrane dynamics, and ciliogenesis.

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Abstract

Ciliogenesis is a highly ordered process that requires membrane trafficking, fusion, and maturation. In this study, we investigated EXOC6A, a component of the exocyst complex known for secretory vesicle trafficking and fusion, and found that it interacts with myosin-Va (Myo-Va) during ciliogenesis. EXOC6A colocalizes with Myo-Va at various stages of ciliogenesis, including preciliary vesicles, ciliary vesicles (CVs), and ciliary sheath membrane during ciliogenesis. We found that EXOC6A vesicles are actively recruited, integrated, and exit from the CVs and the ciliary sheath, implying that EXOC6A vesicles may facilitate continuous cilia membrane remodeling during ciliogenesis. Importantly, EXOC6A knockout impairs ciliogenesis, arresting most cells at the CV stage and preventing recruitment of NPHP and MKS module components to the transition zone. Furthermore, EXOC6A vesicles are transported to the mother centriole via a dynein-, microtubule-, and actin-dependent mechanism. Our results suggest that EXOC6A functions in both early and late stages of ciliogenesis and is involved in orchestrating vesicle dynamics, cilia membrane remodeling, and formation.

Introduction

Most mammalian cells build primary cilia, which act as chemosensors or mechanosensors and as transducers that regulate key developmental signaling pathways. Primary cilia are composed of nine doublet microtubules arranged in a ring and surrounded by a ciliary membrane that is continuous with the plasma membrane (Dawe et al., 2007 ; Nigg and Raff, 2009 ). Primary cilia protrude from the cell surface to perform a wide range of biological functions. Defects in ciliogenesis can lead to a number of genetic disorders known as ciliopathies, which are characterized by loss of vision, disturbed kidney function, obesity, and defects in brain development (Sattar and Gleeson, 2011 ; Valente et al., 2014 .

Ciliogenesis is a highly ordered and complex process. Currently, two distinct pathways have been reported, with different cell types involved: intracellular and extracellular. Polarized epithelial cells use the extracellular pathway, whereas fibroblasts and mesenchymal cells use the intracellular pathway (Breslow and Holland, 2019 ). In the intracellular pathway, myosin-Va

(Myo-Va) facilitates the transportation of preciliary vesicles (PCVs) to the distal appendages (DAs) of the mother centrioles (Wu et al., 2018 [DOI](#)). These PCVs may be derived from the Golgi apparatus (Schmidt et al., 2012 [DOI](#)) and subsequently fuse with a ciliary vesicle (CV) through the action of membrane fusion regulators EHD1 and EHD3 (Lu et al., 2015 [DOI](#)).

This step is critical for the conversion of the mother centriole into a basal body, which is responsible for removing the centriole cap protein, CP110, and recruiting ciliary transition zone (TZ) proteins (Lu et al., 2015 [DOI](#)). As a result, an axoneme extends within the larger CV, which gradually stretches out to form the double membranes of the ciliary shaft and sheath that surround the extending axoneme. Finally, the ciliary sheath (the outer membrane) fuses with the plasma membrane and the axoneme, enveloped by the ciliary shaft membrane, extends from the cell membrane, and makes contact with the extracellular environment (Ghossoub et al., 2011 [DOI](#)). A signaling cascade including RAB11, RABIN8, and RAB8 has also been reported to regulate the ciliary membrane assembly (Chiba et al., 2013 [DOI](#); Feng et al., 2012 [DOI](#); Knödler et al., 2010 [DOI](#); Nachury et al., 2007 [DOI](#); Westlake et al., 2011 [DOI](#); Yoshimura et al., 2007 [DOI](#)). However, the processes of formation of the ciliary sheath and ciliary shaft membrane during ciliogenesis remain unclear.

The exocyst is a multi-subunit protein complex that was first identified in yeast, which mediates the tethering of secretory vesicles to the plasma membrane (Mei and Guo, 2018 [DOI](#)). The complex is composed of eight subunits, named Sec (for secretion; Sec3, Sec5, Sec6, Sec8, Sec10, Sec15, Exo70, and Exo84) in yeast or Exoc (for exocyst-related; EXOC1 to 8) in humans (TerBush et al., 1996 [DOI](#); TerBush and Novick, 1995 [DOI](#)). The exocyst complex is implicated in various cellular processes, including exocytosis, ciliogenesis, cytokinesis, autophagy, cell polarity, migration, tumorigenesis, and fusion of secretory vesicles (Mei and Guo, 2018 [DOI](#)).

Mutations in genes encoding components of the exocyst complex have been associated with various human genetic disorders. For example, a mutation in *EXOC8* has been identified in Joubert syndrome (Dixon-Salazar et al., 2012 [DOI](#)), while a mutation in *EXOC4* has been reported in Meckel-Gruber syndrome (Shaheen et al., 2013 [DOI](#)). Both of these disorders involve clinically and genetically heterogeneous ciliopathy. Additionally, mutations in *EXOC2* cause severe defects in human brain development (Van Bergen et al., 2020 [DOI](#)), while mutations in *EXOC7* and *EXOC8* result in a novel disorder of cerebral cortex development, characterized by brain atrophy, seizures, developmental delay, and microcephaly (Coulter et al., 2020 [DOI](#)). However, the essential roles of these genes in ciliogenesis and their correlation with human genetic disorders are currently unknown.

It has been reported that some components of the exocyst complex are involved in ciliogenesis. For example, Sec10 (EXOC5) was shown to be required for ciliogenesis and cyst formation in vitro, and Sec10 knockdown leads to reduce ciliogenesis (Zuo et al., 2009 [DOI](#); Zuo et al., 2019 [DOI](#)). Furthermore, other subunits such as EXOC3 and EXOC4 have been detected at the cilium with unknown function (Seixas et al., 2016 [DOI](#)). Interestingly, depletion of Sec8, Exo70, or Sec10 does not impair ciliogenesis (Rivera-Molina et al., 2021 [DOI](#)), whereas knockdown of Sec15a (EXOC6A) significantly perturbs ciliogenesis (Rivera-Molina et al., 2021 [DOI](#)). These findings imply that different exocyst subunits may have distinct, non-redundant functions during ciliogenesis. However, the precise localization of EXOC6A within cilia and its specific role in regulating ciliary membrane formation or trafficking events remain largely unknown.

Previously, we demonstrated that the earliest event in ciliogenesis is the transportation of PCVs by Myo-Va to the distal appendages of the mother centriole, and we showed that Myo-Va is released from the ciliary membrane upon fusion of the ciliary sheath with the plasma membrane (Wu et al., 2018 [DOI](#)). However, the mechanism of the assembly of double ciliary membranes (ciliary sheath and shaft) during intracellular ciliogenesis was left unknown. In this study, we provide evidence that the exocyst component EXOC6A interacts with Myo-Va and plays an essential role in ciliary membrane assembly at different stages during ciliogenesis.

Results

Colocalization patterns of EXOC6A and Myo-Va at preciliary vesicles, ciliary vesicles, and ciliary sheath during intracellular ciliogenesis

To investigate the localization of EXOC6A during ciliogenesis, we treated retinal pigment epithelial (RPE1)-based mCherry-tagged Myo-Va cargo binding domain (Myo-Va-GTD)-inducible cells with doxycycline (Dox) for 24 h and then serum starved them for 2 h to induce ciliogenesis. In this Tet-On system, Dox treatment triggers the inducible expression of the mCherry-Myo-Va-GTD transgene. We used antibodies against endogenous EXOC6A and glutamylated tubulin (Glu-tub), a marker of centrioles and ciliary axoneme, for confocal immunofluorescence analysis. Our results showed that the EXOC6A signal was mainly detected as a dot-shaped structure resembling a ciliary vesicle (CV) at the distal end of the mother centriole (Fig. 1A [↗](#), left), while some EXOC6A signals appeared to be colocalized with mCherry-tagged Myo-Va-GTD at the ciliary membrane (Fig. 1A [↗](#), right). To determine the subcellular localization of EXOC6A, we performed immunofluorescence staining using antibodies against CEP164 (a distal appendage marker, (Graser et al., 2007 [↗](#))) and CEP120 (a centriole marker, (Lin et al., 2013 [↗](#))). The PCV/CV/ciliary sheath can be visualized with mCherry-tagged Myo-Va-GTD (Wu et al., 2018 [↗](#)). The cells were then analyzed using three-dimensional structured illumination microscopy (3D-SIM, Fig. 1B [↗](#)) or ultrastructure expansion microscopy (U-ExM, Fig. 1C [↗](#)). Our results showed that EXOC6A signals colocalized with mCherry-Myo-Va-GTD at CV (Fig. 1B and C [↗](#)). To further define the position of the EXOC6A signal, we performed Correlative Light and Electron Microscopy (CLEM) in RPE1-based GFP-EXOC6A-inducible cells (Fig. 1D [↗](#)). Our CLEM results showed that the GFP-EXOC6A signals appeared to localize at PCVs (or vesicles), CVs (Fig. 1D-i-ii [↗](#)), and the elongating ciliary membrane (Fig. 1D-iii-iv [↗](#)).

To precisely analyze the subcellular localization of EXOC6A and Myo-Va during the formation of the ciliary sheath and the shaft membrane, we used 3D-SIM to analyze the RPE1 cells that exogenously express inducible GFP-EXOC6A using known markers combined for the ciliary sheath (Myo-Va-GTD; (Wu et al., 2018 [↗](#))), the ciliary shaft (ARL13B, INPP5E; (Feng et al., 2012 [↗](#); Garcia-Gonzalo et al., 2015 [↗](#); Wu et al., 2018 [↗](#))), and the axoneme (Glu-tub). EXOC6A signals were commonly detected at the Myo-Va-GTD-labeled ciliary sheath (Fig. 1E-i [↗](#), ii), which surrounds the ciliary shaft membrane (labeled ARL13B and INPP5E, Fig. 1E-i [↗](#), iii) and the axoneme (Glu-tub, Fig. 1E-ii, iii [↗](#)). Since our 3D-SIM analysis could not distinguish the spatial distances between INPP5E and the Glu-tub-labeled axoneme due to the limits of its resolution power (100–130 nm), we thus conclude that EXOC6A localizes at the ciliary sheath membrane surrounding the ciliary shaft. Altogether, our data show that EXOC6A, like Myo-Va, localizes at the PCVs, the CVs, and the ciliary sheath membrane.

Spatial and temporal localization of Myo-Va and EXOC6A during ciliogenesis

To investigate the temporal and spatial correlation between the ciliary localization of EXOC6A and Myo-Va during ciliogenesis, we used RPE1-based mCherry-Myo-Va-GTD-inducible cells that were serum starved, released at different time points, and analyzed using 3D-SIM with antibodies against endogenous EXOC6A and Glu-tub (Fig. 2A [↗](#)). We identified five different phenotypes based on the immunostaining patterns of Myo-Va and EXOC6A during cilium assembly (Fig. 2B [↗](#)), which are quantified in Fig. 2C [↗](#). Before serum starvation, a number of EXOC6A-associated vesicles were detected close to the mother centriole (Fig. 2A [↗](#), 0 min, type 1). During the progression of serum starvation, EXOC6A-associated vesicles appeared to gradually accumulate near the mother centriole (Fig. 2A [↗](#), 15 min, type 2) in a process we called “clustering-to-centriole,” and subsequently, EXOC6A signals were mainly detected at the Myo-Va-labeled CVs (Fig. 2A [↗](#), 30 min–1 h, type 3) and the ciliary sheath membrane, surrounding the Glu-tub-labeled axoneme (Fig. 2A [↗](#), 2 h, type 4). In some cells, a tubule-like structure (EXOC6A⁺/Myo-Va⁺) linked to

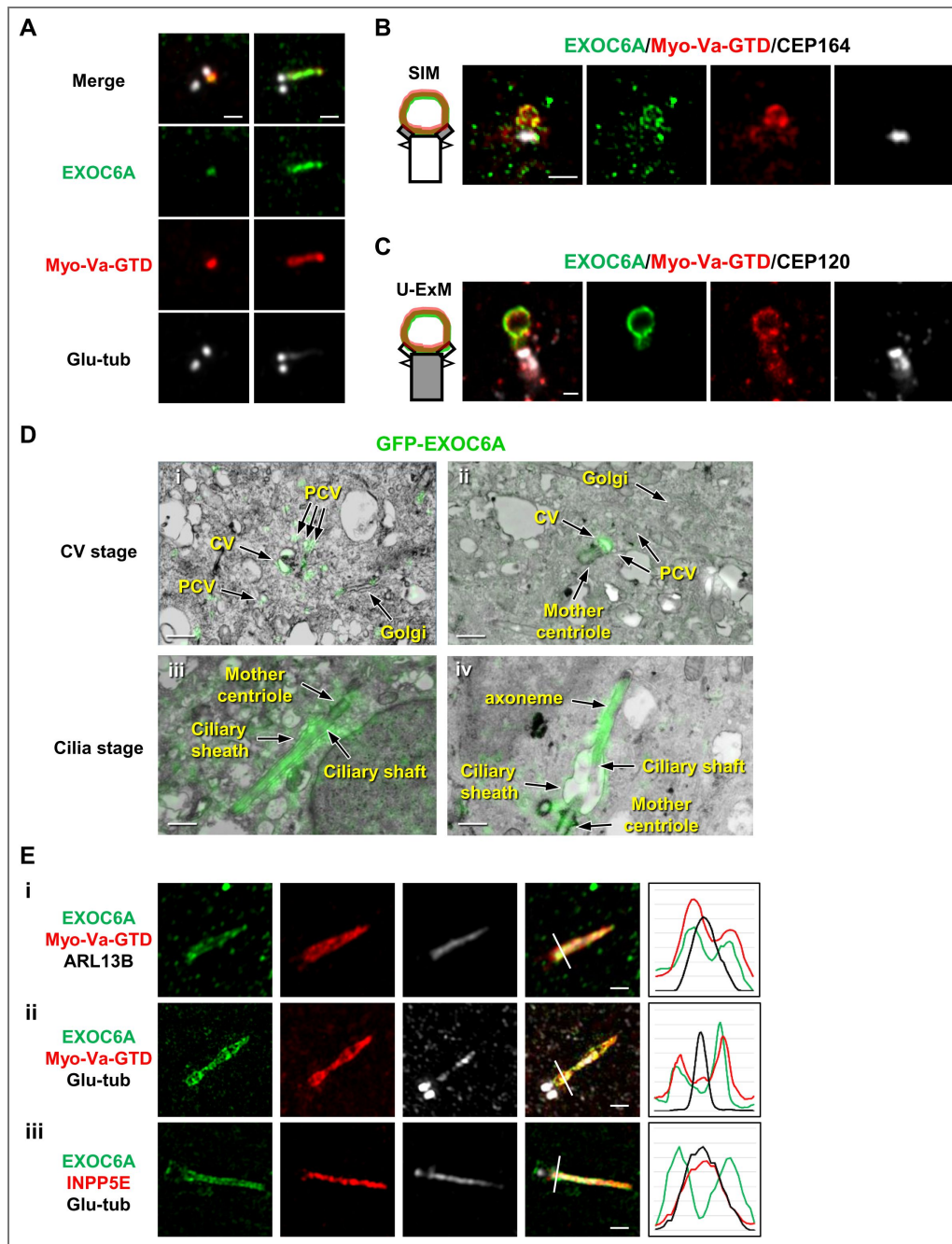


Figure 1. Colocalization patterns of EXOC6A and myosin-Va at preciliary vesicles, ciliary vesicles, and ciliary sheath during ciliogenesis.

(A) RPE1-based mCherry-Myo-Va-GTD cells were treated with Dox for 24 h, followed by serum starvation for 2 h. Cells were then stained with antibodies against EXOC6A (green) and polyglutamylated tubulin (Glu-tub) (gray) and analyzed using immunofluorescence confocal microscopy. (B and C) RPE1-based mCherry-Myo-Va-GTD-inducible cells were treated as described in A and immunostained with antibodies against EXOC6A (green), CEP120 (centriole marker), and CEP164 (distal appendage marker, gray). Images were captured using 3D-SIM or ultra-expansion microscopy (U-ExM) with an LSM880 confocal system. (D) CLEM analysis of the localization of GFP-EXOC6A during ciliogenesis. RPE1-based GFP-EXOC6A-inducible cells were treated with Dox for 24 h and subjected to serum starvation for 2 or 24 h to observe the CVs or cilia membrane, respectively. Images were taken via SIM and TEM and then merged based on their relative localization. GFP-EXOC6A signals were located at PCVs, CVs (i-ii), and the ciliary membrane (iii-iv). (E) RPE1-based GFP-EXOC6A and mCherry-Myo-Va-GTD-inducible cells were treated with Dox for 24 h and analyzed via 3D-SIM using the indicated antibodies. Right panels show fluorescence profile plots. EXOC6A colocalized with myosin-Va at the ciliary sheath, while ARL13B (i) and INPP5E (iii) are the ciliary shaft markers. Scale bars are 1 μ m.

the ciliary sheath membrane was observed (Fig. 2A, 4 h, white arrow). Finally, both EXOC6A and Myo-Va were released from the ciliary sheath membrane when it fused to the plasma membrane, while the Glu-tub-labeled axoneme continued to protrude into the extracellular milieu (Fig. 2A, 24 h, type 5). Interestingly, we observed that only a portion, but not all of the EXOC6A-labeled vesicles colocalized with Myo-Va-labeled PCVs (Fig. 2A, 0 and 15 min). However, once the CVs formed, both EXOC6A and Myo-Va-GTD signals were detected at the CVs (Fig. 2A, 30 min, 45 min, and 1 h). We present a schematic model in Figure 2B that shows the localization of EXOC6A (green) and mCherry-Myo-Va-GTD (red) during ciliogenesis and the types of immunostaining patterns.

To further understand the relationship between the clustering-to-centriole progression of EXOC6A and Myo-Va during the early stage of ciliogenesis, we analyzed the intensity of both proteins clustering to the centrioles and calculated their overlapping co-efficiency correlation R (Pearson correlation coefficient) using Zeiss Zen software (Fig. S1). We observed a gradual increase in the mean intensity of EXOC6A and Myo-Va clustering to the centrioles from 0 to 30 min (Fig. S1A–C). Additionally, the correlation R between EXOC6A and Myo-Va significantly increased from 0 to 30 min (Fig. S1D). These results suggest a high correlation between Myo-Va-labeled vesicles and EXOC6A-labeled vesicles during the progression of clustering-to-centriole.

FRAP analysis of the dynamic localization of Myo-Va- and EXOC6A-labeled vesicles during ciliogenesis

We found that some EXOC6A structures had a tubule-like structure extending from the ciliary sheath (Fig. 2A, 4 h, type 4), suggesting the dynamic assembly of the ciliary membrane during ciliogenesis. To gain further insights into the dynamic relationship between Myo-Va- and EXOC6A-labeled vesicles during early CV formation and the later sheath membrane assembly stage, we performed live cell imaging with fluorescence recovery after photobleaching (FRAP). FRAP is a widely used technique to measure the dynamics of ciliary membrane protein localization (Kee et al., 2012; Westlake et al., 2011). After photobleaching at CVs or the ciliary sheath region, the EXOC6A signals were recurrently recruited to the CVs (Fig. 3A, Video 1) and the ciliary sheath membrane within seconds (Fig. 3B, Video 2). Together, our data indicated that the EXOC6A-labeled vesicles are continuously recruited to and fused with the ciliary membrane not only at the CVs but also at the ciliary sheath. To further investigate the dynamic properties of the ciliary membrane, we performed live-cell super-resolution imaging using the ELYRA 7 SIM system. We used SiR-tubulin, a fluorescent probe derived from the microtubule-stabilizing drug docetaxel, to label endogenous microtubules. This allowed us to visualize the structure of centrioles and axonemes during ciliogenesis in live imaging experiments (Fig. S2). Our results show that EXOC6A co-localizes with Myo-Va-GTD on the ciliary membrane and surrounds the SiR-tubulin-labeled axoneme (Fig. S2A and S2B). In addition, we detected a dynamic tubule-like structure that extended and retracted from the ciliary membrane (Fig. S2A, S2C, and Video 3). To better observe vesicle trafficking and membrane dynamics, we used the single-channel and burst modes of the ELYRA 7 SIM system to achieve higher temporal resolution in live imaging. Our results showed that GFP-EXOC6A-labeled vesicles gradually moved to and fused with CVs (Fig. 3C, white arrow; Video 4) and the ciliary pocket, a membrane domain found at the base of primary cilia (Fig. 3D, white arrow, Video 5). Interestingly, almost at the same time, some GFP-EXOC6A-labeled vesicles were fused with or secreted from the ciliary pocket (Fig. 3D, red arrow, Video 5). Unexpectedly, we observed that some GFP-EXOC6A-associated tubular structures (Fig. 3D, yellow arrow) or GFP-EXOC6A-labeled vesicles were released directly from the ciliary membrane rather than from the ciliary pocket (Fig. 3D, orange arrow). Together, our results indicate that GFP-EXOC6A-associated vesicles/membranes are highly dynamic and continuously recruited to and/or released from CVs during the early stages of ciliogenesis, and that they are also fused with or excreted from the ciliary membrane during later stages of ciliogenesis. We, therefore, propose that EXOC6A vesicles are responsible for the exchange of materials within existing CVs and the ciliary sheath during ciliogenesis, facilitating the addition or removal of membranes or membrane proteins.

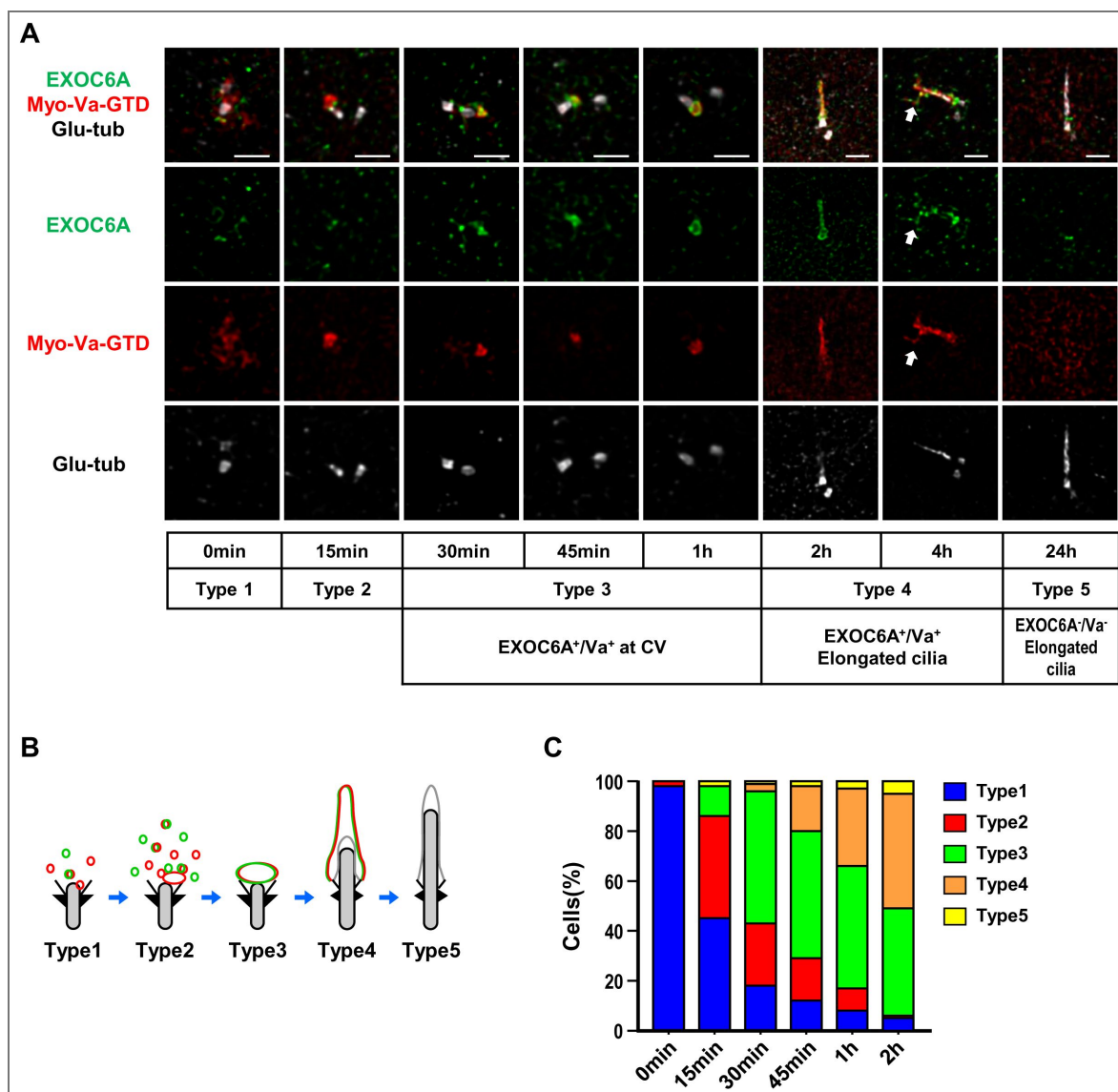


Figure 2. Spatial-temporal localization of EXOC6A and Myo-Va during ciliogenesis.

(A) RPE1-based mCherry-Myo-Va-GTD-inducible cells were treated with Dox for 24 h, serum starved, released at different time points, and analyzed via 3D-SIM using the indicated antibodies. The white arrow indicates a tubule extending from the ciliary membrane. (B, C) Schematic model showing the localization patterns of EXOC6A (green) and Myo-Va (red) during ciliogenesis, and the percentages of cells with different types of staining patterns (%) are shown in C. Scale bars are 1 μ m.

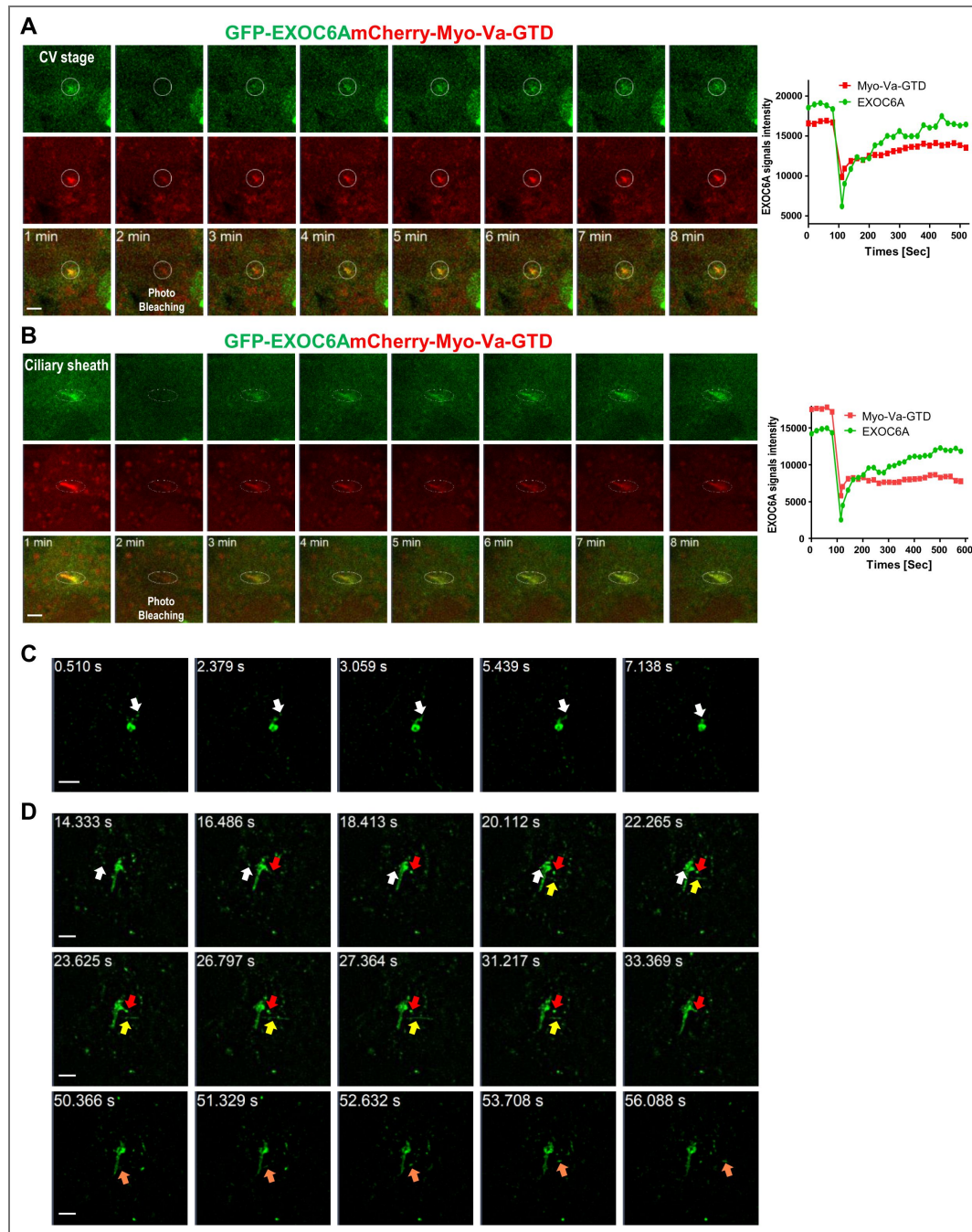


Figure 3. FRAP analysis of Myo-Va- and EXOC6A-labeled vesicles during ciliogenesis.

(A and B) RPE1-based GFP-EXOC6A and mCherry-Myo-Va-GTD-inducible cells were treated with Dox for 24 h and serum starved for 30 min. Regions of CVs (A) or the ciliary membrane (B) of the cells were photobleached using the LSM880 confocal microscope with the 405 nm laser (related to Videos 1 and 2). Signal intensity of EXOC6A is shown in the right panel. (C and D) Dynamic localization of GFP-EXOC6A at CVs (C) or the ciliary membrane (D) during ciliogenesis. RPE1-based GFP-EXOC6A-inducible cells were treated with Dox for 24 h and serum starved for 30 min. Images were taken using an Elyra 7 high-speed live-cell imaging system. Series of images in (C) shows the progression of fusion of GFP-EXOC6A-associated vesicles into CVs (white arrow). Series of images in (D) shows the process by which GFP-EXOC6A-associated vesicles fuse into the ciliary pocket (white arrow) and simultaneously exit or reintegrate into the ciliary pocket (red arrow). Some images also show the release of tubule-like structures in the ciliary membrane (yellow arrow), or the excretion of vesicles in the ciliary membrane (orange arrow). These data suggested that the components of CVs, the ciliary pocket, and the ciliary membrane are highly dynamic (related to Videos 4 and 5). Scale bars are 2 μ m.

Myo-Va and EHD1 are required for EXOC6A-labeled CV formation, and EXOC6A is associated with Myo-Va

We previously reported that Myo-Va-mediated transportation of PCVs to the mother centriole is the earliest event that defines the onset of ciliogenesis (Wu et al., 2018). In this study, we aimed to examine the sequential order of PCV transportation mediated by Myo-Va and EXOC6A during ciliogenesis in both RPE1-based *Myo-Va* knockout (KO) and *EXOC6A* KO cells. We constructed two independent *EXOC6A* knockout (KO) cell lines using the CRISPR–Cas9 system. The success of the gene disruption was validated at the genomic level through target site sequencing (Fig. S3A), a schematic diagram of predicted peptide products (Fig. S3B), and Western blot analysis (Fig. S3C), confirming the absence of EXOC6A expression in all KO cell lines. All independent KO cell lines exhibited similar ciliary generation defects (Fig. S3D). The EXOC6A signal is located at CV (Fig. S3E) and cilia sheath (Fig. S3F), but is absent in both *EXOC6A* KO cell lines. Our results showed that few EXOC6A-labeled vesicles and no EXOC6A-labeled CVs were detected at the distal end of the mother centriole in the *Myo-Va* KO cells (Fig. 4A). In contrast, Myo-Va-labeled CVs were frequently observed at the mother centrioles of *EXOC6A* KO cells (Fig. 4B). Similar results were also observed in the other *EXOC6A* KO cell line. Together, our findings suggest that EXOC6A is not essential for the formation of Myo-Va-labeled CVs during ciliogenesis.

EHD1 was reported to be essential for the fusion of PCVs to form larger CVs during the early stage of ciliogenesis (Lu et al., 2015). To investigate the sequential order of EHD1- and EXOC6A-mediated ciliary membrane assembly, we depleted EHD1 expression using siRNA in RPE1 cells. Our results showed that EHD1 depletion led to a reduction in the formation of EXOC6A-labeled CVs (Fig. 4C). In contrast, *EXOC6A* KO did not appear to affect EHD1-mediated fusion of PCVs to form a large EHD1-labeled CV at the mother centriole (Fig. 4D). Together, our findings suggest that EHD1 is required for EXOC6A-labeled CV formation, while EXOC6A is not essential for the formation of EHD1-labeled CVs. Additionally, previous studies reported that Myo2p, the yeast homolog of human Myo-Va, interacts with sec15, the yeast homolog of human EXOC6A (Jin et al., 2011). The present study showed that some, but not all of the EXOC6A-labeled PCVs colocalized with Myo-Va-labeled PCVs (Fig. 2A, 0 and 15 min). We speculate that Myo-Va may transport some EXOC6A-labeled vesicles to the CVs for ciliary membrane assembly. Our co-immunoprecipitation experiments showed that endogenous EXOC6A can co-immunoprecipitate with Myo-Va, supporting this hypothesis (Fig. 4E).

Finally, previous studies have indicated that the removal of centriolar protein CP110 from the distal end of the mother centriole is necessary for axoneme growth (Spektor et al., 2007). It has been proposed that CP110 removal occurs after the docking of CV to the DA of the mother centriole (Lu et al., 2015; Wu et al., 2018). To investigate the impact of EXOC6A loss on CP110 removal, we examined CP110 localization in *EXOC6A* KO cells. Our results showed that CP110 is appropriately removed from the distal end of the mother centriole in *EXOC6A* KO cells, indicating that EXOC6A is not required for this process (Fig. S4), which further support that EXOC6A is not essential for CV formation. Taken together, our findings suggest that Myo-Va transports a subset of EXOC6A-labeled vesicles to the existing CVs of the mother centriole for ciliary membrane assembly. However, EXOC6A is not essential for CV formation.

Transportation of EXOC6A-labeled vesicles to the mother centriole is via a dynein-, MT-, and Arp2/3-dependent pathway

Our previous study showed that the transportation of PCVs to the distal appendages of the mother centriole is mediated by Myo-Va, dynein, microtubules (MTs), and the Arp2/3 complex (Wu et al., 2018). In the current study, we found that Myo-Va and EHD1 are required for EXOC6A-labeled CV formation and that Myo-Va co-immunoprecipitates with EXOC6A (Fig. 4). It has been reported that dynein is necessary for the correct positioning of the Golgi complex near the centrosome upon serum starvation (Palmer et al., 2009; Wu et al., 2018), and that the

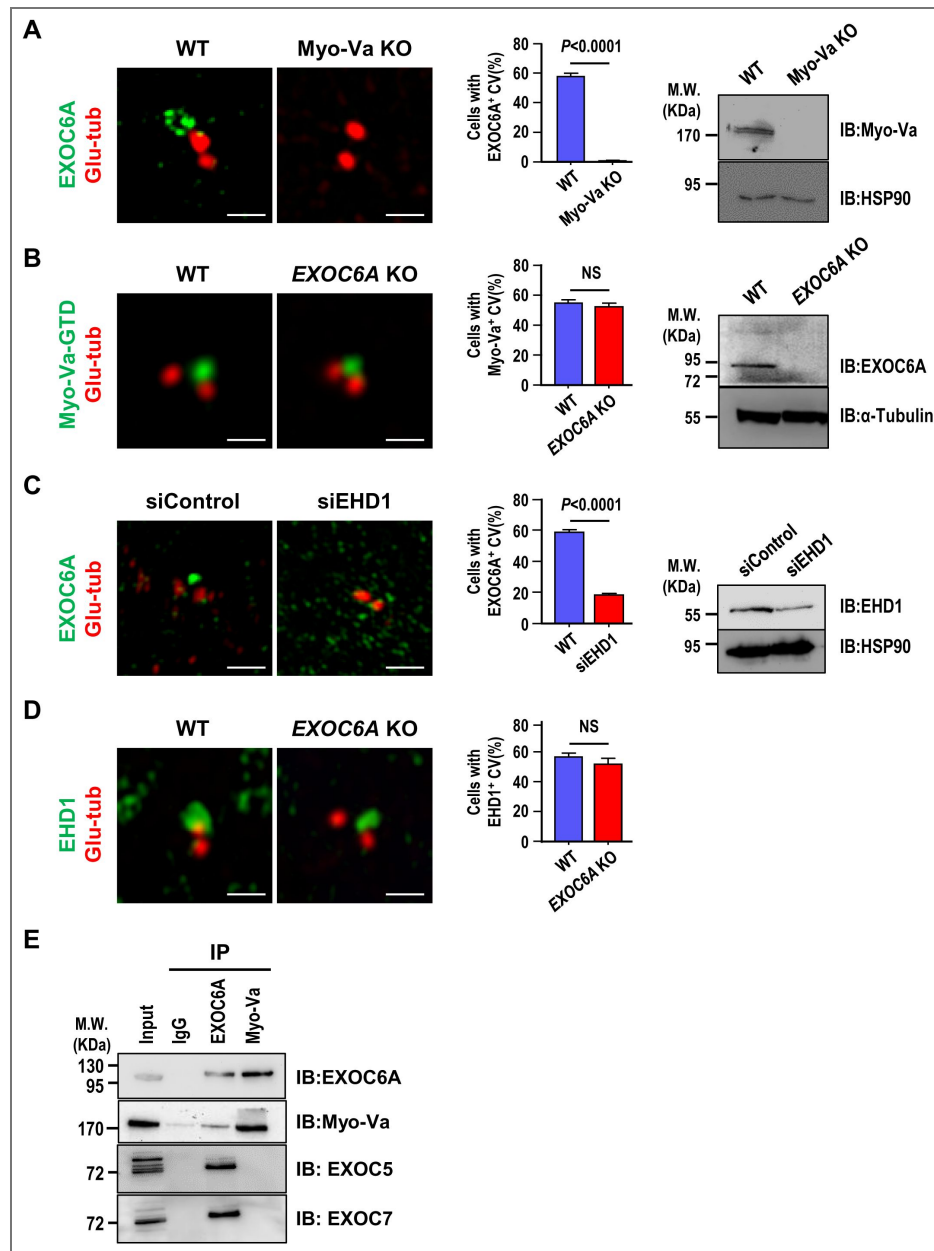


Figure 4. Myo-Va and EHD1 are required for EXOC6A-labeled CV formation, and EXOC6A is associated with Myo-Va.

(A) Myo-Va is required for EXOC6A-labeled CV formation. RPE1-based *Myo-Va* KO cells and WT cells immunostained with antibodies against EXOC6A (green) and Glu-tub (red). Quantification of cells with EXOC6A-positive CVs (middle panel) and immunoblotting results (right panel) are shown. (B) EXOC6A is not required for Myo-Va-labeled CV formation. RPE1-based mCherry-Myo-Va-GTD-inducible cells against *EXOC6A* KO background and WT cells were immunostained with antibodies against Glu-tub (red). For easy differentiation, the mCherry-Myo-Va-GTD signal is converted to green. Quantification of cells with Myo-Va-associated CVs (middle panel) and immunoblotting results (right panel) are shown. (C) EHD1 is required for EXOC6A-labeled CV formation. RPE1 cells treated with siControl or siEHD1 for 48 h were immunostained with antibodies against EXOC6A (green) and Glu-tub (red). Quantification of cells with EXOC6A-positive CVs (middle panel) and immunoblotting results (right panel) are shown. (D) EXOC6A is not required for EHD1-labeled CV formation. RPE1-based *EXOC6A* KO and WT cells were immunostained with antibodies against EHD1 (green) and Glu-tub (red). Quantification of cells with EHD1-positive CVS is shown (right panel). (E) Co-IP experiments analyzed the association between EXOC6A and its potential binding proteins. Cell lysates were immunoprecipitated (IP) with EXOC6A or Myo-Va antibodies, followed by immunoblotting with the indicated antibodies. Error bars in A–D represent mean $\pm$ s.d. from at least 3 independent experiments with 100 cells per experiment. *P*-value was determined with two-tailed Student's *t*-test. *P* < 0.05 was considered statistically significant. NS, not significant. Scale bars are 1 μ m.

inhibition of dynein activity with ciliobrevin D suppresses ciliogenesis (Firestone et al., 2012; Wu et al., 2018). Therefore, we investigated whether the transport of EXOC6A-associated vesicles to the DA of the mother centriole also requires dynein, MTs, and the Arp2/3 complex.

RPE1 cells were transfected with siRNAs targeting Golgin160, a protein required for the recruitment of dynein to the Golgi complex (Yadav et al., 2012), or treated with ciliobrevin D, a known cytoplasmic motor dynein inhibitor that suppresses ciliogenesis (Firestone et al., 2012), following the protocol described in Figure 5A. Our findings showed that depletion of Golgin160 led to a decrease in EXOC6A-labeled CV signals at or near the mother centriole (Fig. 5B, D) and resulted in the dispersion of Golgi matrix protein 130 (GM-130)-labeled Golgi complexes (Fig. 5B, E), ultimately leading to a reduction in the total number of cilia (Fig. 5F). Similar results were also observed in RPE1 cells treated with 10 μ M ciliobrevin D (Fig. 5C-i, D-F). We next treated RPE1 cells with 10 μ M nocodazole (NZ, an MT-depolymerizing drug) and examined its effect on ciliogenesis. Our results showed that the EXOC6A-labeled CVs and the cilia number are greatly reduced in NZ-treated cells (Fig. 5C-ii, D, and F). These data suggest that a subset of EXOC6A-positive vesicles may originate from the Golgi apparatus. Furthermore, the accumulation of these vesicles at the mother centrioles is highly sensitive to disruption of dynein or microtubules, suggesting that efficient transport of these vesicles may depend on the integrity of the microtubule network. However, more experiments are required to confirm this conclusion.

Since Myo-Va-mediated transport of PCVs to the mother centrioles relies on an Arp2/3-dependent branched actin network surrounding mother centrioles (Wu et al., 2018), we next treated RPE1 cells with siARP2 (Fig. 5G). A reduced cilia number (Fig. 5G and H) and reduced EXOC6A-labeled CVs (Fig. 5G and I) were observed in siARP2-treated cells. A similar result was also found in cells treated with 200 μ M CK666 (a known specific Arp2/3 inhibitor) or high-dose (10 μ M) cytochalasin D (CytoD), which has been reported to reduce the association of ARP2 with the branched actin network (Fig. 5H-J). It has been reported that low concentrations (100 nM–1 μ M) of CytoD can promote ciliogenesis and increase cilia length (Cao et al., 2023; Kim et al., 2015; Kim et al., 2010; Wu et al., 2018), while high concentrations (10 μ M) lead to a significant inhibition of cilia formation (Wu et al., 2018). To further investigate the dose-dependent effects of actin cytoskeleton rearrangement on ciliogenesis, we treated RPE1 cells with CytoD at concentrations ranging from 200 nM to 10 μ M. Immunofluorescence staining of cilia using ARL13B (green) and Glu-tubulin (red) antibodies showed that low concentrations of CytoD (200 nM–2 μ M) resulted in significant cilia elongation, whereas higher doses (4 μ M, 7 μ M and 10 μ M) gradually inhibited cilia formation (Fig. S5). Consistent with our previous findings (Wu et al., 2018), our data showed that cilia length and cilia ratio increased at low doses of CytoD (200 nM–2 μ M) but gradually decreased with increasing doses (7 μ M and 10 μ M, Fig. S5B and C).

Taken together, our current studies reinforce our previous view (Wu et al., 2018) that low-dose of CytoD treatment (200 nM–2 μ M) can stabilize and/or enhance the formation of Arp2-associated branched actin filaments. However when cells were treated with high doses of CytoD (7 μ M and 10 μ M, Fig. S5) or the specific Arp2/3 inhibitor, CK-666 (Fig. 5), the association of Arp2 with the branched actin network was significantly reduced, thereby inhibiting ciliogenesis. In conclusion, our data suggest that Myo-Va carries EXOC6A vesicles to the DA of mother centriole via a dynein-, microtubule-, and actin-dependent pathway.

Loss of EXOC6A inhibits ciliogenesis, and some cells exhibit abnormal ciliary morphology when passing through the CV block

We found that EXOC6A signals are detected at CVs and the ciliary sheath (Fig. 1 and 2). We also found that EXOC6A-associated vesicles are dynamically recruited to or excreted from CVs and/or the ciliary membrane (Fig. 3). We next examined the effects of EXOC6A deletion on ciliogenesis. As shown in Figure 6, EXOC6A KO significantly inhibits cilia assembly, resulting in either no cilia or dotted CV signals after serum starvation (SS, 30 min–48 h). Upon serum starvation for 30 min, Myo-Va-GTD-labeled CVs were detected at the mother centriole in both wild-type control (Fig. 6A) and EXOC6A KO cells (Fig. 6A'). In control cells, the Myo-Va-GTD signal was mainly detected at the ciliary sheath membrane (Fig. 6A, SS 2h) 2 h after serum starvation,

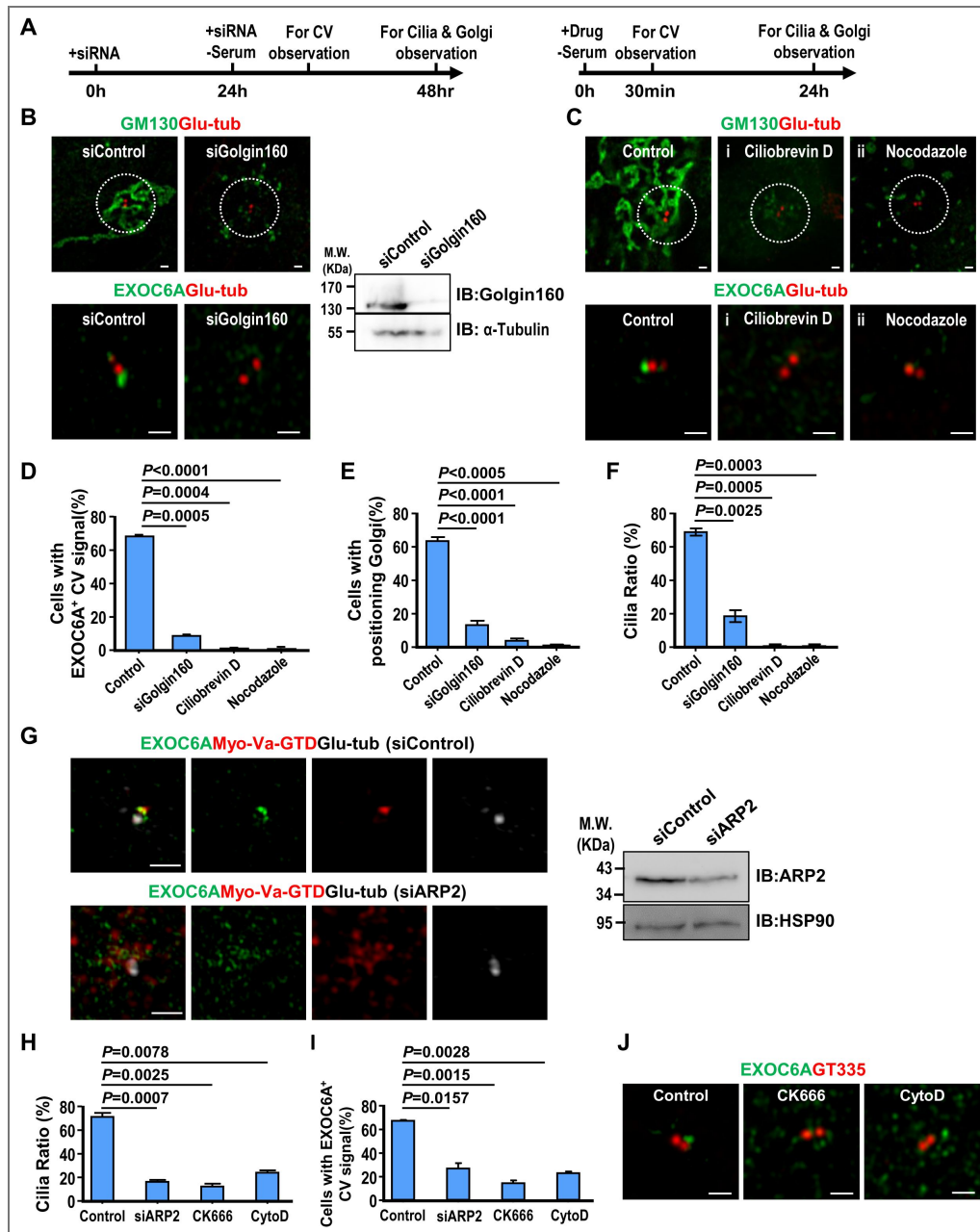


Figure 5. Transportation of EXOC6A-associated vesicles to the mother-centriole occurs via a dynein-, actin-, and MT-dependent pathway

(A) Schematic showing the protocols of siRNA (left) or drug (right) treatment. (B) RPE1 cells were treated with siControl or siGolgin160 and analyzed via confocal fluorescence microscopy or immunoblotting using the indicated antibodies. (C) RPE1 cells were treated with 10 μ M ciliobrevin D (i) or 10 μ M nocodazole (ii) and analyzed via confocal fluorescence microscopy using the indicated antibodies. (D-F) Cells with EXOC6A-associated CVs (D), a positioned Golgi (E), or a cilia ratio (F) were analyzed via 3D-SIM and quantified. Cells with a positioned Golgi were defined as those showing all GM130-labeled Golgi signals concentrated within a 7 μ m radius surrounding the Glu-tub-labeled centrioles. (G) RPE1-based mCherry-mycosin-Va-GTD-inducible cells were treated with siARP2 as described in A (left) and analyzed via confocal microscopy or immunoblotting using the indicated antibodies. (H, I) Quantifications of the cilia ratio (H) and EXOC6A-associated CVs (I) is shown. (J) RPE1 cells were treated with 200 μ M CK666 or high-dose (10 μ M) CytoD as described in A (right panel) and analyzed via confocal microscopy using the indicated antibodies. Quantification of the cilia ratio (H) and EXOC6A-associated CVs (I) is shown. * in (B, right panel) indicates non-specific bands. Error bars represent mean $\pm$ s.d. from at least 3 independent experiments with 100 randomly selected cells. P -value was determined with two-tailed Student's t -test. $P < 0.05$ was considered statistically significant. NS, not significant. Scale bars are 1 μ m.

while the Myo-Va-GTD signals dissociated from the ciliary sheath membrane after longer starvation (SS 24 h or SS 48 h). Interestingly, most Myo-Va-GTD signals in *EXOC6A* KO cells were present as dotted CVs after longer starvation (SS 48 h), suggesting that ciliogenesis is arrested at the CV stage (Fig. 6A [↗](#)).

We next used an antibody against ARL13B (a cilia shaft marker) to label the ciliary membrane and found that most *EXOC6A* KO cells exhibited no cilia (~45%, Fig. 6C [↗](#)) or dotted ARL13B signals (~50%) 2 h after SS (Fig. 6B [↗](#); Fig. 6C [↗](#), SS 2 h), a pattern similar to Myo-Va staining (Fig. 6A [↗](#)). Unexpectedly, after prolonged serum starvation, we could still detect approximately ~15% (SS 24 h) to ~30% (SS 48 h) of *EXOC6A* KO cells carrying ARL13B-labeled cilia protruding from the basal body. However, these ARL13B-labeled cilia showed a low intensity of ARL13B and Glu-tub signals (Fig. 6D and E [↗](#)) and the ARL13B-labeled cilia membrane exhibited a fragmented and punctuated staining pattern (enhanced in Fig. 6B [↗](#), SS 48 h in lower panel). Our findings suggest that *EXOC6A* may play a crucial role in the further maturation of CVs into the ciliary sheath/shaft membrane during ciliogenesis.

To examine whether *EXOC6A* is required for ciliogenesis and whether its function can be rescued, we analyzed cilia formation in wild-type (WT), *EXOC6A* KO, and *EXOC6A* KO cells exogenously expressing GFP-*EXOC6A*. Immunofluorescence staining of cells with ARL13B and Glu-tubulin antibodies showed a significant reduction in the number of cilia in *EXOC6A* KO cells. Importantly, exogenous expression of GFP-*EXOC6A* in the KO cells resulted in a statistically significant recovery of the ciliated cell population ($P=0.0041$) and a concomitant reduction in stalled vesicles, confirming that this phenotype was caused by *EXOC6A* loss. (Fig. S6 [↗](#)). Electron microscopy was further used to investigate the cilia morphology in the *EXOC6A* KO cells after prolonged serum starvation (72 h). To examine ciliary structure in *EXOC6A* KO cells in more detail, we used a 72 h-serum starvation protocol, which provided sufficient time for cilia assembly. Our results showed a perturbation in cilia assembly in *EXOC6A* KO cells (Fig. 6F and G [↗](#)). When compared with the wild-type control (Fig. 6F-i [↗](#) and ii), cilia in *EXOC6A* KO cells exhibited some ciliogenic defects, such that ciliogenesis was often arrested at the DA/PCV or CV stage (Fig. 6F-iii and iv [↗](#)), and some exhibited morphologically curved cilia bending in the TZ region (Fig. 6F-vi [↗](#), 4/37 in Fig. 6G [↗](#)). Intriguingly, a small proportion of KO cells produced an elongated large CV in the absence of an extended axoneme (Fig. 6F-v [↗](#), 4/37 in Fig. 6G [↗](#)). Consistent with this finding, curved cilia (Fig. 6B [↗](#), SS 24 h, lower panel) and a low intensity of Glu-tub-labeled axoneme were observed in *EXOC6A* KO cells via confocal immunofluorescence microscopy (Fig. 6B [↗](#), SS 48 h, lower panel. Fig. 6E [↗](#)). Nonetheless, we still observed 3 out of 37 morphologically normal-like cilia in *EXOC6A* KO cells (Fig. 6F-vii [↗](#), 6G). Taken together, our data suggest that depletion of *EXOC6A* primarily affects cells exhibiting no CV (showing only DA/PCV; Fig. 6F-iii [↗](#), Fig. 6G [↗](#)) and/or arrests cells at the CV stage, while once it passes through CV stage after prolonged serum starvation, it may interfere with subsequent cilia membrane formation at later stages of ciliogenesis.

***EXOC6A* is required to recruit NPHP and MKS module components to the TZ and GPR161 and BBS9 to the ciliary membrane**

Our EM study revealed both bending cilia at the TZ (4/37, Fig. 6G-vi [↗](#)) and morphologically normal-like cilia (3/37, Fig. 6G-vii [↗](#)). Intriguingly, immunofluorescence analysis revealed that the ARL13B signal intensity was greatly reduced in the normal-like cilia membrane in *EXOC6A* KO cells (Fig. 6D [↗](#)), suggesting a possible defect in the TZ of cilia. The TZ is a diffusion barrier at the ciliary membrane that acts as a gate to regulate the entry and exit of ciliary proteins required for signal transduction (Jensen and Leroux, 2017 [↗](#)). Previous studies have identified three conserved protein modules (MKS, NPHP, and CEP290), composed of numerous protein complexes that cooperate in the assembly and gating function of the TZ (Gonçalves and Pelletier, 2017 [↗](#)). CEP290, located at a proximal position near the mother centriole, acts as a hub that interacts with the NPHP and MKS module complexes (Garcia-Gonzalo and Reiter, 2017 [↗](#)).

To verify the integrity of these abnormal cilia in *EXOC6A* KO cells, we examined the recruitment of several known TZ proteins, including CEP290, NPHP, and MKS module proteins. Our results showed that the recruitment of CEP290 to the ciliary base was normal in all examined ciliated

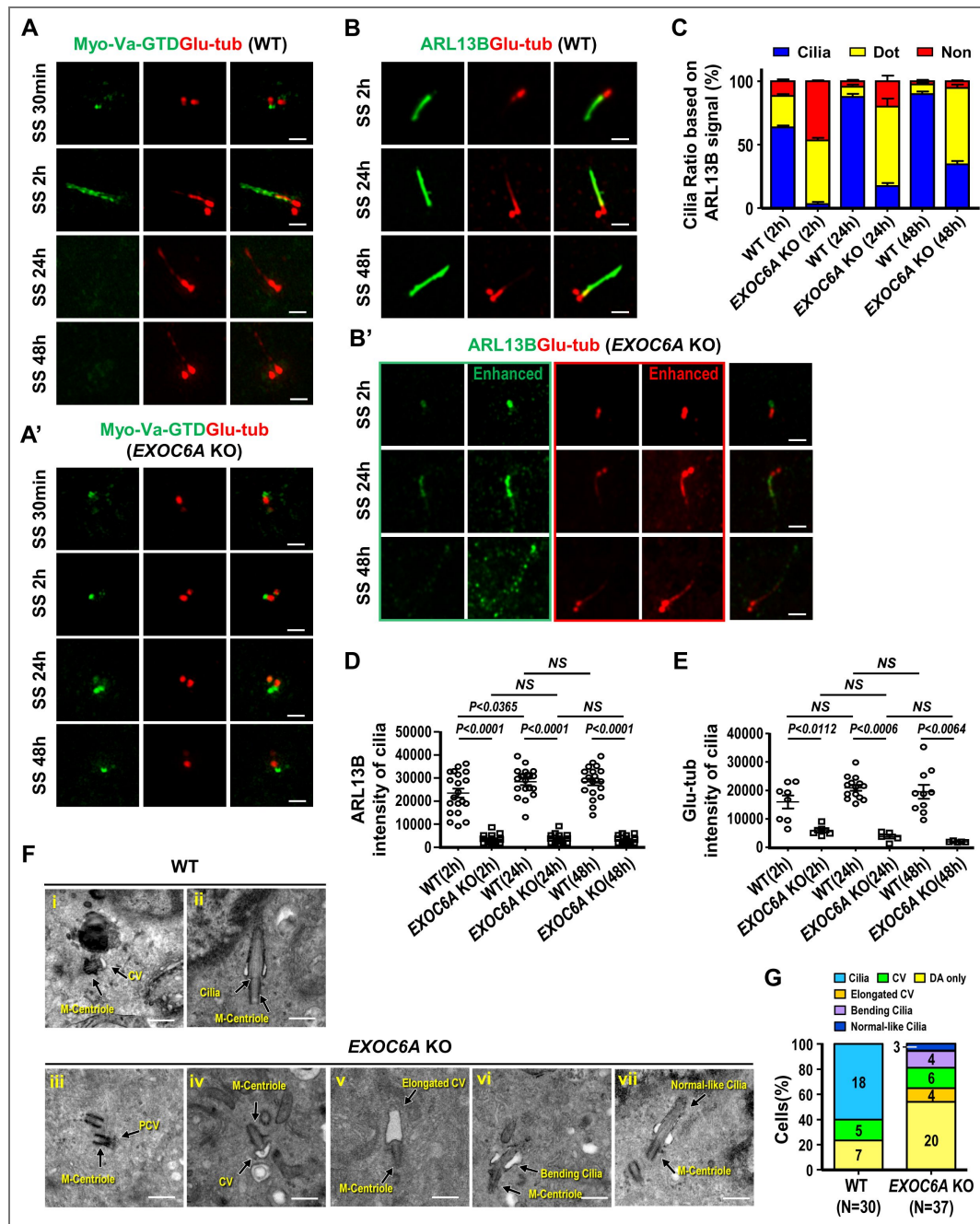


Figure 6. Loss of EXOC6A inhibits ciliogenesis, and some cells exhibit an abnormal ciliary morphology when passing through the CV block.

(A-C) RPE1-based mCherry-Myo-Va-GTD-inducible WT (A and B) and EXOC6A KO (A' and B') cells were treated with Dox for 24 h, serum starved (SS) at different time points, and analyzed via confocal fluorescence microscopy with the indicated antibodies. mCherry-Myo-Va-GTD signal is artificially converted to green for discrimination. (B) RPE1 WT and EXOC6A KO cells were serum starved at different time points and analyzed via confocal fluorescence microscopy with the indicated antibodies. For comparison, the intensities of ARL13B and Glu-tub of EXOC6A KO cells with non-enhanced and enhanced signals are shown in B'. (C) Cilia ratio measured through ARL13B labeling (non-enhanced signal). (D and E) Quantitation of non-enhanced ARL13B (D) and non-enhanced Glu-tub intensity on cilia (E) of WT or EXOC6A KO cells is shown. (F) WT and EXOC6A KO cells were serum starved for 72 h. Morphologies of normal or abnormal cilia, including DA, CVs, and ciliary membranes, were examined via EM (F) and quantified (G; N is the number of cells examined). Error bars in C, D, and E represent mean $\pm$ s.d. from at least 3 independent experiments from 100 randomly selected cells. *P*-value was determined with two-tailed Student's *t*-test. *P* < 0.05 was considered statistically significant. NS, not significant. Scale bars are 1 μ m in A and B. Scale bars are 500 nm in F.

cells, indicating that EXOC6A deletion did not affect CEP290 targeting to the TZ (Fig. 7A). However, the recruitment of several MKS complex proteins (TCTN2, AHI1, MKS3, and MKS1) to the basal body (mother centriole) was severely impaired in *EXOC6A* KO ciliated cells (Fig. 7B), while NPHP8 recruitment was only partially affected (Fig. 7B). Collectively, our findings suggest that EXOC6A is not required for CEP290 targeting to the TZ but plays an essential role in the recruitment of several NPHP and MKS module proteins to the basal body, which potentially affects the gating function of the TZ in *EXOC6A* KO cells.

We next investigate whether the EXOC6A deficiency affects the transport of ciliary membrane proteins, we examined the localization of GPR161 and BBS9 in *EXOC6A* KO cells. GPR161 is a G-protein-coupled receptor that negatively regulates the Sonic hedgehog (Shh) signaling pathway (Mukhopadhyay et al., 2013), while BBS9 is a core component of the BBSome complex and is essential for cilia homeostasis (Wingfield et al., 2018). Our results showed that both GPR161 and BBS9 were properly recruited to the cilia in WT cells (Fig. 7C). However, in *EXOC6A* KO cells, the entry of these proteins into the ciliary compartment was completely blocked (Fig. 7C). Our findings suggest that EXOC6A is essential for the recruitment of a subset of TZ proteins and membrane proteins to the cilia during ciliogenesis.

Discussion

Cilia assembly is a complex, multistep process involving the accumulation and attachment of PCVs at the distal appendage, CV formation, and the subsequent extension of CVs into the ciliary sheath, alongside the formation of the ciliary shaft, ultimately leading to the mature cilia structure (Breslow and Holland, 2019; Long and Huang, 2019; Nachury et al., 2010). Currently, our understanding of the molecular mechanisms governing these steps remains incomplete. Exocytosis is a process where intracellular secretory vesicles fuse with the plasma membrane, aiding in delivering integral membrane proteins to the cell surface and releasing materials into the extracellular space (Martin-Urdiroz et al., 2016). Human EXOC6A exhibits a high degree of similarity to the product of the *Saccharomyces cerevisiae* *SEC15* gene. In yeast, SEC15 is essential for the movement of vesicles from the Golgi apparatus to the cell surface (Bowser and Novick, 1991). Previous studies have indicated that SEC15 interacts with Myo2p (the yeast homolog of human Myo-Va) and plays a role in vesicle exocytosis (Donovan and Bretscher, 2012; Donovan and Bretscher, 2015). Recently, exocyst was found to mediate the recycling of internal cilia and depletion of EXOC6A/Sec15a, but not other exocyst components, leads to a reduction in ciliogenesis (Rivera-Molina et al., 2021). This finding suggests that EXOC6A may have a unique function for ciliogenesis. However, the relationship and underlying mechanism between EXOC6A and Myo-Va in the context of ciliogenesis remain unclear.

In this study, we elucidated the role of the exocyst component EXOC6A in ciliogenesis and produced evidence for the model by which Myo-Va mediates trafficking of EXOC6A vesicles to the DA or existing CV of mother centriole in the early stages and to the ciliary membrane in the later stages of ciliogenesis (Fig. 8). However, further investigation will be required to fully establish the proposed hierarchy and molecular details. In this model, we found that a subset of EXOC6A-positive signals colocalized with Myo-Va-labeled signals during the early phase of ciliogenesis, and that they exhibited a high correlation during clustering-to-centriole progression after initiating ciliogenesis (Fig. S1A-D). Intriguingly, not all EXOC6A-labeled signals were positive for Myo-Va (Fig. 2A), suggesting the presence of at least two populations of Myo-Va-labeled vesicles in cells: one that is EXOC6A+/Myo-Va+ and another that is EXOC6A-/Myo-Va+ (Fig. 2A and Fig. 8).

In this model, ciliogenesis begins shortly after serum-starvation-induced quiescence. Initially, Myo-Va mediates the transport of PCVs (EXOC6A-/Myo-Va+ and EXOC6A+/Myo-Va+ vesicles) to the DA of the mother centriole, followed by the formation of PCVs to a CV via EHD1 (Lu et al., 2015; Wu et al., 2018). However, EXOC6A is not absolutely required for CV formation, as *EXOC6A* KO did not interfere with the formation of Myo-Va- or EHD1-labelled CV (Fig. 4B, 4D). Once the CV is formed, the transport of EXOC6A vesicles to the existing CV in the early stages of ciliogenesis is still mediated by Myo-Va (Fig. 4A). The above process involves a two-step transportation mechanism: initially, dynein mediates the movement of Myo-Va-associated vesicles (either EXOC6A-positive or

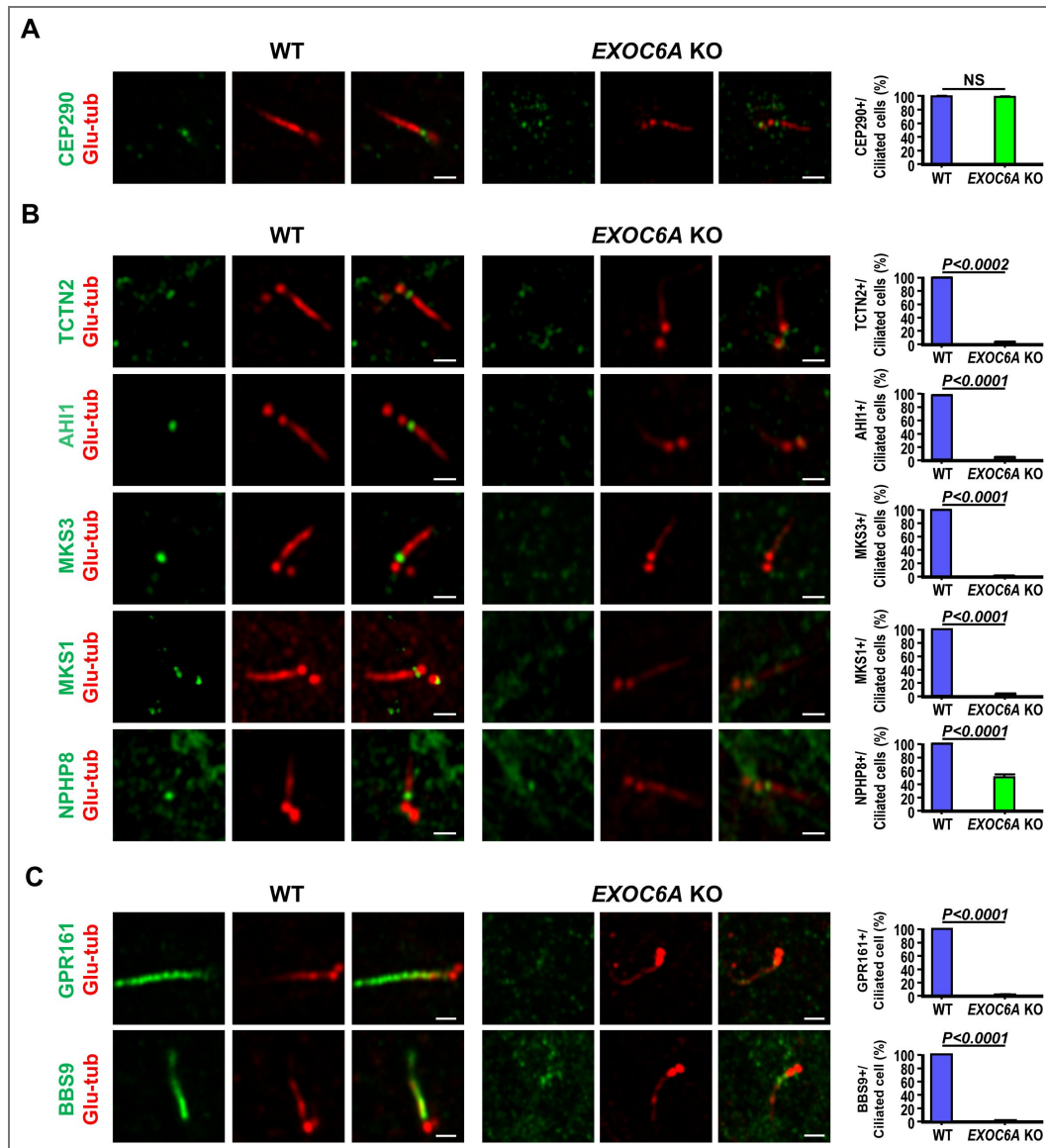


Figure 7. EXOC6A is required for the recruitment of a subset of TZ proteins and ciliary membrane proteins to the cilia

(A) WT and EXOC6A KO cells were fixed for 2 h (A) or 24 h (B, C) after serum starvation and analyzed via confocal fluorescence microscopy using indicated antibodies. Glu-tub staining labeled the centriole. Quantification of the fluorescence signals of various MKS proteins, NPHP8, and ciliary membrane proteins (GPR161 and BBS9) is shown in the right panel. Error bars represent the mean $\pm$ s.d. from at least 3 independent experiments with 100 randomly selected cells per experiment. P -value was determined with two-tailed Student's t -test. $P < 0.05$ was considered statistically significant. NS, not significant. Scale bars, 1 μ m.

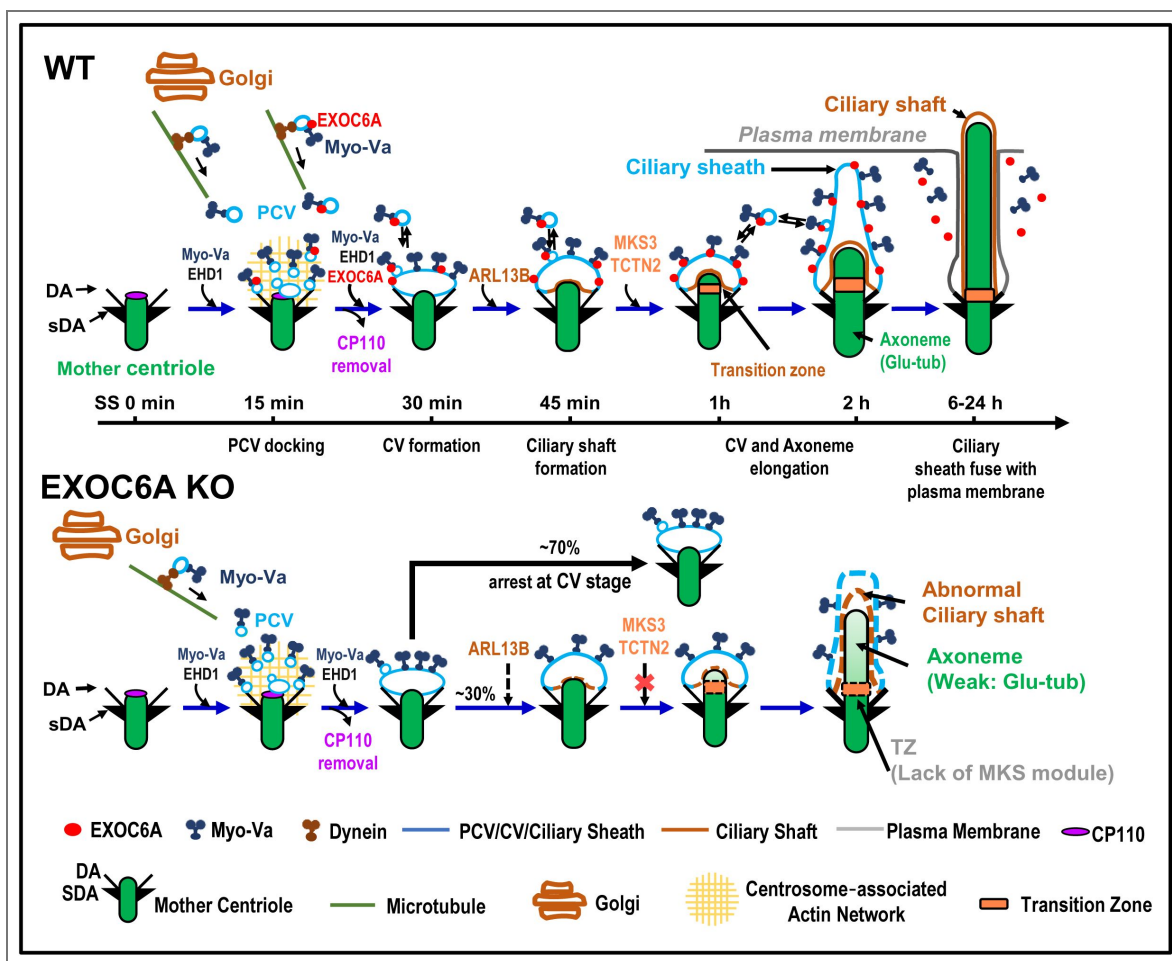


Figure 8. Ciliogenesis model and comparison between WT and *EXOC6A* KO cells.

EXOC6A is located at the PCV/CV/ciliary sheath, and Myo-Va mediated the transportation of *EXOC6A* vesicles to the mother centriole via a dynein-, microtubule-, and actin-dependent pathway. *EXOC6A*-associated vesicles are continuously recruited and fused or excreted from the CVs or ciliary membrane. Depletion of *EXOC6A* impairs ciliogenesis, and most cells are arrested at the CV stage. There is a lack of MKS module proteins of the TZ in cilia with defects in *EXOC6A* KO cells. PCV, preciliary vesicle; CV, ciliary vesicle; DA, distal appendage; sDA, subdistal appendage.

EXOC6A-negative) towards the pericentrosomal region along microtubules (MTs) (Fig. 5B–F). Subsequently, Myo-Va takes over and drives these vesicles (EXOC6A-positive or -negative) from the pericentrosomal region to the DAs of the mother centriole. This process is accomplished through the utilization of the ARP2/3-associated branched actin network (Fig. 5G–I). Notably, both of these Myo-Va-labeled vesicles require the function of EHD1 to form a CV (Fig. 4D; Wu et al., 2018). Our findings that EHD1 depletion (Fig. 4D) and *Myo-Va* KO (Fig. 4A) severely impaired the formation of EXOC6A-positive CVs at the mother centriole, suggest that Myo-Va and EHD1 play a role in promoting the formation of EXOC6-positive CVs during the early stages of ciliogenesis.

Although EXOC6A is not absolutely required for the initial CV formation (Fig. 4B, 4D), our data revealed another role of EXOC6A in the later stages of ciliogenesis. Our FRAP analysis and live cell imaging showed that a subset of EXOC6A-labeled vesicles displayed high dynamism throughout ciliogenesis, continually being recruited to, fusing into, or exited from existing CVs in the early stages and/or the ciliary membrane (or ciliary pocket) in the later stages of ciliogenesis (Fig. 3). These EXOC6A-positive vesicles, likely derived from the Golgi or recycling endocytic vesicles, are transported to the ciliary membrane (or ciliary pocket) and/or fused with/exited from the ciliary membrane in the later stages of ciliogenesis. The exocyst has been implicated in the transport of vesicles carrying ciliary proteins to the cilia (Wu and Guo, 2015). Thus, our findings suggested a potential role of EXOC6A vesicles, which may supply or remove proteins or lipid moieties from the CVs and/or the ciliary membrane (or ciliary pocket) during ciliogenesis.

Additionally, our results showed that *EXOC6A* KO led to the majority of the cells being arrested at the CV stage or exhibiting no or few cilia (Fig. 6C); only a small proportion of cilia were able to elongate, exhibiting an abnormal ciliary morphology (Fig. 6B, 6F). Electron microscopic analysis of these abnormal cilia revealed elongated CV (lacked axoneme, Fig. 6F–v), curved cilia bending at the TZ region (Fig. 6F–vi), or normal-like cilia (Figure 6B–vii, whose structure and membrane composition need to be further characterized). Further confocal immunofluorescence analysis revealed a curved ciliary morphology (Fig. 6B, SS24h) and impaired recruitment of several known TZ proteins, including TCTN2, AHI1, MKS3, and MKS1, to the basal body in *EXOC6A* KO cells (Fig. 7). Taken together, our findings suggest that EXOC6A is required for the early and the later stages of ciliogenesis as providing or removing ciliary components (proteins or membrane moieties) to the existing CV or ciliary membrane, which highlights its multifaceted involvement in the process of ciliary membrane formation.

Furthermore, the Rab11–Rabin8–Rab8 signaling pathway, which involves the recruitment of Rabin8 (a GDP–GTP exchange factor for Rab8) and subsequent activation of Rab8, has been implicated in ciliary membrane assembly (Knödler et al., 2010; Nachury et al., 2007; Westlake et al., 2011; Yoshimura et al., 2007). Recently, RAB34, located at the ciliary sheath (Ganga et al., 2021; Stuck et al., 2021), was also reported to be required for ciliogenesis (Xu et al., 2018). The relationship between EXOC6A, Rab11 signaling, and RAB34 in ciliogenesis is not clear. Future experiments dissecting the underlying mechanisms between Rab11 signaling, RAB34, and EXOC6A could provide a more complete picture of ciliary membrane assembly during ciliogenesis.

Finally, ciliopathy-associated genes have been demonstrated to impact various stages of mouse cortical development (Guo et al., 2015), and severe human ciliopathies are linked to developmental abnormalities in the forebrain (Andreu-Cervera et al., 2021). Mutations or deletions in genes encoding exocyst proteins, including *EXOC2*, *EXOC4*, *EXOC7*, and *EXOC8*, have been associated with ciliopathies and neural developmental disorders (Coulter et al., 2020; Dixon-Salazar et al., 2012; Shaheen et al., 2013; Turkyilmaz et al., 2021; Van Bergen et al., 2020). Interestingly, deletions involving the 5' portion of the *EXOC6A* gene and two adjacent cytochrome p450 genes (*CYP26A1* and *CYP26C1*) are associated with autosomal-dominant nonsyndromic optic aplasia (ONA), an extremely rare disorder that causes unilateral or bilateral blindness in the affected eye (Meire et al., 2011). The molecular basis remains unclear. Further studies are needed to explore the specific roles of various types of ciliary vesicles, the underlying mechanisms of their trafficking/fusion, and their relevance to ciliopathies and neural developmental disorders.

In summary, we have identified a novel role for EXOC6A in ciliogenesis and determined that it is essential for both early and late stages of this process. EXOC6A vesicles may transport proteins or lipid moieties to and from the CV (early phase) and the ciliary pocket/ciliary membrane (late phase) during ciliogenesis. However, EXOC6A is not absolutely necessary for the initial CV formation; its function may be to help assemble the complete transition zone and promote the maturation of existing CVs into the ciliary sheath/shaft membrane. Furthermore, transportation of EXOC6A vesicles to the mother centriole is mediated by Myo-Va and occurs via a dynein-, MT-, and Arp2/3-dependent pathway. Our study provides new insights into the function of EXOC6A in cilia membrane assembly during ciliogenesis. Future studies are needed to explore the specific roles of different types of ciliary vesicles, the underlying mechanisms of their trafficking and fusion, and their relevance to ciliopathies and neural developmental disorders.

Methods

Plasmids and antibodies

cDNAs encoding full-length EXOC6A were obtained via RT-PCR using total RNAs from human HEK293T cells and subcloned in-frame into pcDNA4/TO/myc-His-A (Invitrogen) or pLVX-Tight-Puro (BD Biosciences Clontech) GFP expression vectors. GFP- and mCherry-myosin-Va-GTD constructs have been previously described (Wu et al., 2018 [DOI](#)). The sequences of all constructed plasmids were confirmed.

Commercial antibodies used were anti-Sec15 (clone [15S2G6], 1/100 for IF, Kerafast), anti-EXOC6 (1/100 for IF and 1/2000 for WB, Novus), anti-EXOC5 (1/2000 for WB, Proteintech), anti-EXOC7 (1/2000 for WB, Proteintech), anti-polyglutamylated tubulin (Glu-tub, GT335, 1/100 for IF, AdipoGen), anti-CEP164 (1/500 for IF, Novus), anti-GFP (Abcam and Clontech), anti-ARL13B (1/600 for IF, Proteintech), anti-myosin-Va (1/500 for IF and 1/5000 for WB, Novus), anti-EHD1 (1/500 for IF and 1/5000 for WB, Novus), anti-GM130 (1/1000 for IF, Abcam), anti-CP110 (1/600 for IF, Proteintech), anti-Golgin160 (1/2000 for WB, Novus), anti-Arp2 (1/1000 for WB, Abcam), anti-TMEM67 (MKS3, 1/400 for IF, Proteintech), anti-TCTN2 (1/400 for IF, Proteintech), anti-AHI1 (1/400 for IF, Proteintech), anti-MKS1 (1/400 for IF, Proteintech), anti-BBS9 (1/400 for IF, Proteintech), and anti-GPR161 (1/400 for IF, Proteintech). The secondary antibodies were Alexa 488/568/647 conjugated anti-mouse or rabbit (Invitrogen). The rabbit polyclonal antibody against CEP120 (residues 639–986) was as described in our previous paper (Lin et al., 2013 [DOI](#); Tsai et al., 2019 [DOI](#)).

Cell culture and cell lines

Human telomerase-immortalized retinal pigment epithelial cells (RPE1) (Wu et al., 2018 [DOI](#)) originally obtained from ATCC were maintained in DMEM/F12 (1:1) supplemented with 10% FBS. Cells were transfected with various cDNA constructs using TransIT-LT1 Transfection Reagent (Mirus). All cell lines were tested for mycoplasma contamination and found negative.

The gRNA expression plasmids were obtained from Addgene (Addgene plasmid # 48138; <http://n2t.net/addgene:48138> [DOI](#); RRID:Addgene_48138 [DOI](#)), having been generated by inserting annealed primers into the gRNA cloning vector pSpCas9-2A-GFP (PX458) (Ran et al., 2013 [DOI](#)). The targeting sequences for the EXOC6A gRNA (gRNA1:5'-TTGTATCCGTAATCATGACA -3', gRNA2:5'-AAGTACTGATACCAACCGA -3' and gRNA3:5'-ACAAGTTTAACTCACCAGGAAGG -3') primer pair purchased from Mission Biotech were annealed and cloned into PX458. For transfection, 2.5 µg gRNA (PX458 plasmid) was transfected with LT-1 reagent (Mirus) into RPE1 cells, according to the manufacturer's instructions. The KO-1 clone was generated using a single-guide RNA strategy (Fig. S3A-a [DOI](#)). The PX458 vector encoding gRNA1, targeting exon 2 of the EXOC6A gene, was transfected into RPE1 cells. The KO-2 independent clone was generated using a dual-guide RNA strategy to induce a wider range of deletion (Fig. S3A-b [DOI](#)). Vectors encoding gRNA2 and gRNA3, targeting the flanking regions of exon 3 to exon 4, were co-transfected into cells. The transfected cells were sorted into single cells (96-well plates) using a cell sorter (FACSARIA, BD Biosciences) based on GFP

signal and then amplified. To confirm the depletion of EXOC6A, genomic DNAs isolated from *EXOC6A* knockout colonies were subjected to PCR using the following primers: and 5'-3' and 5'-GC TTCAGAAAAGAGAATACTCCT -3' for *EXOC6A* KO-1(gRNA1) and 5'-TTGGGTCAGTGATTGAATTG -3' and 5'-CCAAATAATCTGTAATCCATA -3' for *EXOC6A* KO-2 (gRNA 2 and 3). The sequences of PCR products were then confirmed as shown in Fig. S3A [\[4\]](#). Based on sequence analysis, a schematic diagram of the predicted amino acid sequence is shown in Fig. S3B [\[4\]](#). Also, the protein expression of expanded colonies was confirmed by Western blotting (Fig. S3C [\[4\]](#)) and IF (Fig. S3E [\[4\]](#) and F).

RPE1-based doxycycline (DOX)-inducible cell lines were as described previously (Wu et al., 2018 [\[4\]](#)). To obtain RPE1 cell lines that inducibly expressed GFP-EXOC6A, GFP-Myo-Va-GTD, and mCherry-Myo-Va-GTD, lentiviruses generated using the target cDNAs in the pLVX-tight-puro vector were used to infect RPE1 Tet-On cells (stably expressing rtTA, Clontech). The infected cells were sterile-sorted using a cell sorter (FACSARIA, BD Biosciences) for GFP or mCherry, and the positive cells were selected and expanded as inducible cell lines.

The siRNAs and non-targeting siRNA control were transfected into cells using Lipofectamine RNAiMAX (Invitrogen), according to the manufacturer's protocol. To increase knockdown efficiency, two rounds of silencing were used before the experiments. To ensure the specificity of gene knockdown and minimize off-target effects, we used siRNA sequences that have been validated in our previous studies (Wu et al., 2018 [\[4\]](#)). The efficiency of gene depletion was confirmed by Western blotting. The siRNA sequences for siEHD1 (5'-GGAGAGAUCCUACCAGAAGA-3'), siGolgin-160 (5'-CAGCCUCCUUGGCCGCGAGGGCCUC-3'), and siARP2 (5'-CCAGCUUUGGUUGGAAGACCUAUUA-3') were purchased from Invitrogen.

Immunoprecipitation and Western blotting

RPE1 cells were lysed in RIPA buffer (50 mM Tris-HCl, pH 8.0, 150 mM NaCl, 1% NP-40, 0.5% sodium deoxycholate, 20 mM β -glycerophosphate, 20 mM NaF, 1 mM Na₃VO₄) supplemented with protease inhibitors (1 mg/mL leupeptin, 1 mg/mL pepstatin, and 1 mg/mL aprotinin) for 1 h at 4°C. Lysates were clarified by centrifugation for 10 min at 4°C, and the supernatants were immunoprecipitated with the indicated antibodies for 2 h at 4°C. Protein G Sepharose beads were subsequently added and incubated overnight at 4°C with gentle rotation. The beads were then washed three times with RIPA buffer. Bound proteins were eluted by boiling in SDS sample buffer, separated by SDS-PAGE, and analyzed by Western blotting with the appropriate antibodies.

Immunofluorescence confocal microscopy

Immunofluorescence confocal microscopy was carried out as previously described (Wu et al., 2018 [\[4\]](#)). Briefly, cells were grown on coverslips and fixed in methanol at -20 °C for 20 min. The cells were then blocked with 10% normal goat serum in PBST. After blocking, the cells were incubated with primary antibodies, washed with PBST, and then incubated with Alexa Fluor 488-, 568-, or 647-conjugated secondary antibodies (1/500; Invitrogen). The samples were visualized using an LSM880 Airyscan system (Carl Zeiss). FRAP experiments were performed on live RPE1 cells co-expressing GFP-EXOC6A and mCherry-Myo-Va-GTD using a Zeiss LSM 880 confocal microscope. Time-lapse imaging was conducted with a time interval of 20 seconds. At each time point, Z-stacks consisting of 5 optical sections were captured to cover the entire structure. Following the acquisition of 5 pre-bleach baseline images, the GFP-EXOC6A and mCherry-Myo-Va-GTD signals within a defined region of interest were photo-bleached using the 405 nm laser at 100% intensity for 100 iterations. The fluorescence recovery was monitored over time, and intensity changes were quantified from maximum intensity projections of the Z-stacks using ZEN software.

Super-resolution microscopy (3D-SIM)

The cells were fixed and stained as described (see above), and multi-colored beads (100 nm; Invitrogen) were added before mounting. Sixteen-bit super-resolution images with five phases and five rotations were obtained using a Zeiss ELYRA PS.1 LSM780 system or a Zeiss ELYRA7 system

(Carl Zeiss). We used a plan-apochromatic 63×/1.4 NA oil-immersion M27 objective in combination with the default grid (stripe sim in ELYRA PS.1 and Lattice sim plus SIM square). We used the "optimized" setting for the z-stack interval. The raw images were reconstructed using ZEN Black (Carl Zeiss) under the default parameters. The reconstructed images were corrected with signals of beads for chromatic aberration in the x, y, and z directions, and the images were aligned accordingly with ZEN Black. We followed the standard guidelines (Demmerle et al., 2017 [DOI](#)) and used the SIM-Check software (Ball et al., 2015 [DOI](#)) to verify the raw, reconstructed, and calibrated data. To visualize centriole and axoneme structures during live-cell imaging, we employed SiR-tubulin, a fluorogenic probe that binds to endogenous microtubules (spirochrome).

Ultrastructure expansion microscopy (U-ExM)

U-ExM was used to reveal the fine cellular architecture of the cells. Cells were processed for this U-ExM as previously described (Gambiarotto et al., 2019 [DOI](#)). In brief, cells were fixed with methanol or 4% PFA based on our target. First, cells on coverslips were incubated in a PBS solution containing 0.7% formaldehyde (FA; 36.5–38%; F8775; Sigma-Aldrich) plus 0.15% acrylamide (AA; 40%; A4058; Sigma-Aldrich) for 5 h at 37 °C. Next, the cells on coverslips were incubated in U-ExM monomer solution, composed of 19% (wt/wt) sodium acrylate (SA; 97–99%; 408220; Sigma-Aldrich), 10% (wt/wt) AA, and 0.1% (wt/wt) *N,N'*-methylenebisacrylamide (BIS; 2%; M1533; Sigma-Aldrich) at 37 °C for 1 h in the dark for gelation. We transferred coverslips with gel to 6-well plates, added 1 mL denaturing buffer (200 mM SDS, 200 mM NaCl, and 50 mM Tris in ultrapure water, pH 9), and shook for 15 min at room temperature (RT). We then separated gel from coverslips and transferred the gel to 1.5 mL centrifuge tubes with fresh denaturing buffer, then incubated at 95 °C for 30 min. After denaturation, the gels were placed in beakers filled with ddH₂O overnight for expansion. For antibody staining, the gels were PBS washed and then incubated with primary antibodies for 1 or 2 nights at RT (depending on the antibodies). The gels were washed in PBST (PBS plus 0.1% tween20) and incubated with a secondary antibody for 3 h at 37 °C. The gels expanded by around 4x after this treatment. The samples were then visualized using an LSM880 Airyscan system (Carl Zeiss).

TEM and CLEM

Cells were processed for transmission electron microscopy and CLEM as previously described (Reddick and Alto, 2012 [DOI](#)). Briefly, RPE1 and RPE1-based *EXOC6A* KO cells were grown on Aclar film (Electron Microscopy Sciences) and fixed in 2% glutaraldehyde (GA) containing 1% tannic acid in 0.1 M cacodylate buffer. The fixed cells were post-fixed with 1% OsO₄ in 0.1 M cacodylate buffer at room temperature (RT) for 30 min. Then, the cells were further stained with 1% uranyl acetate at RT for 1 h, dehydrated in a graded series of ethanol, infiltrated, and embedded in Spurr's resin. Ultrathin sections (70 nm) were stained with 4% uranyl acetate (UA) and Reynold's lead citrate for 10 min and examined with an electron microscope (Tecnai G2 Spirit TWIN, FEI).

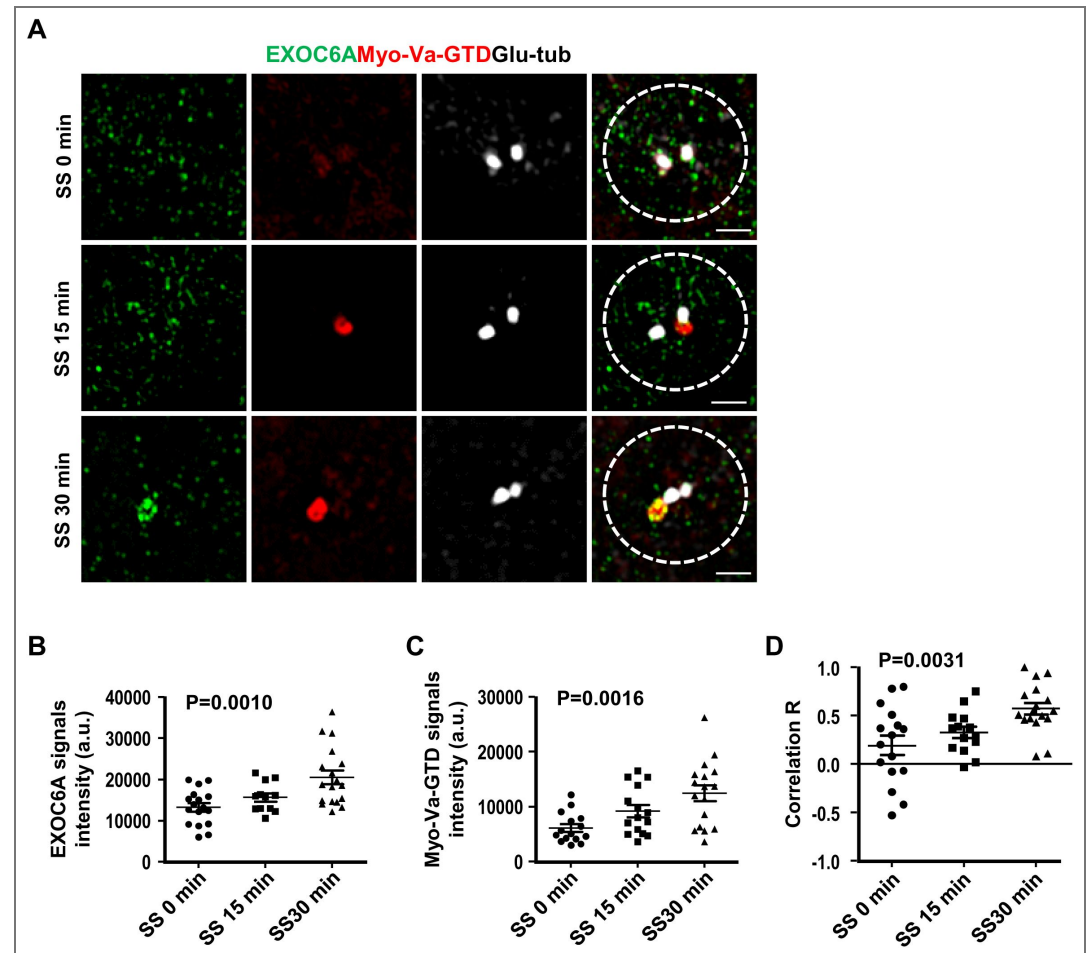
For CLEM, RPE1-based GFP-*EXOC6A*-inducible cells were grown on laser-etched glass gridded coverslips affixed to the bottom of live-cell dishes (MatTek). Cells were treated with DOX for 24 h and serum starved for 2 or 24 h. The cells were then fixed with 2 % glutaraldehyde plus 1% tannic acid in 0.1 M cacodylate buffer, and the GFP-*EXOC6A* signals were imaged via 3D-SIM using a Zeiss ELYRA system (Carl Zeiss). The cells were then embedded in EPON resin. After EPON polymerization, resin blocks were detached from the glass dishes. Using the grid patterns imprinted in the resin, serial thin sections (70 nm) of the squares of interest were cut. The sections were stained with 4% uranyl acetate and Reynold's lead citrate for 10 min, and the cells were examined with an electron microscope (Tecnai G2 Spirit TWIN, FEI). All EM images were processed using Gatan Digital Micrograph software (Gatan).

Statistics and reproducibility

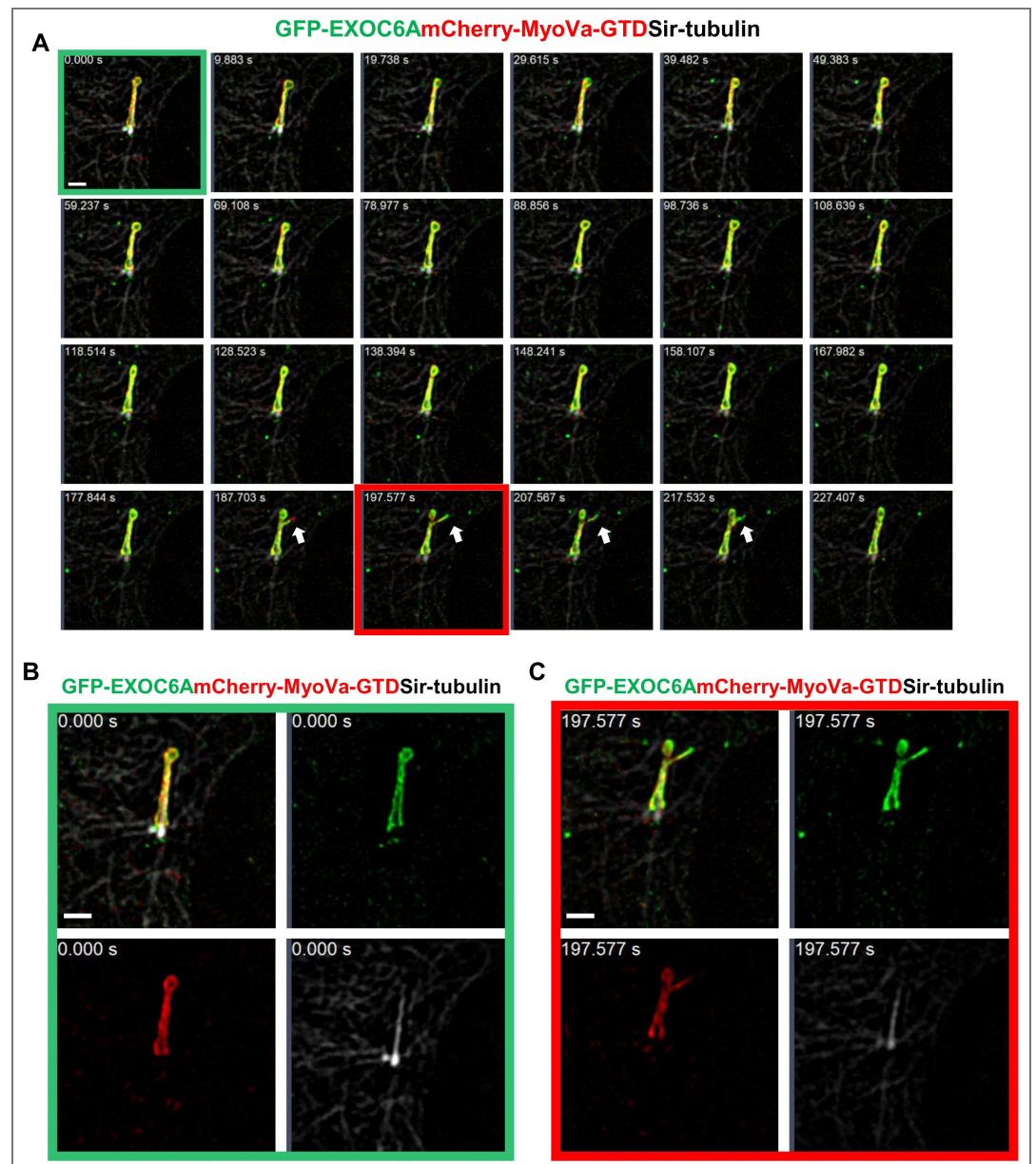
Statistical analyses were performed using GraphPad Prism. Results are presented as mean ± standard derivation (s.d.), as specified in the figure legends. Sample size and the number of repeated experiments are also described in the legends. Statistical differences between two

datasets were analyzed using the two-tailed paired Student's *t*-test. The precise *P*-values are shown in the figures. $P < 0.05$ was considered statistically significant. All experiments were repeated at least three times.

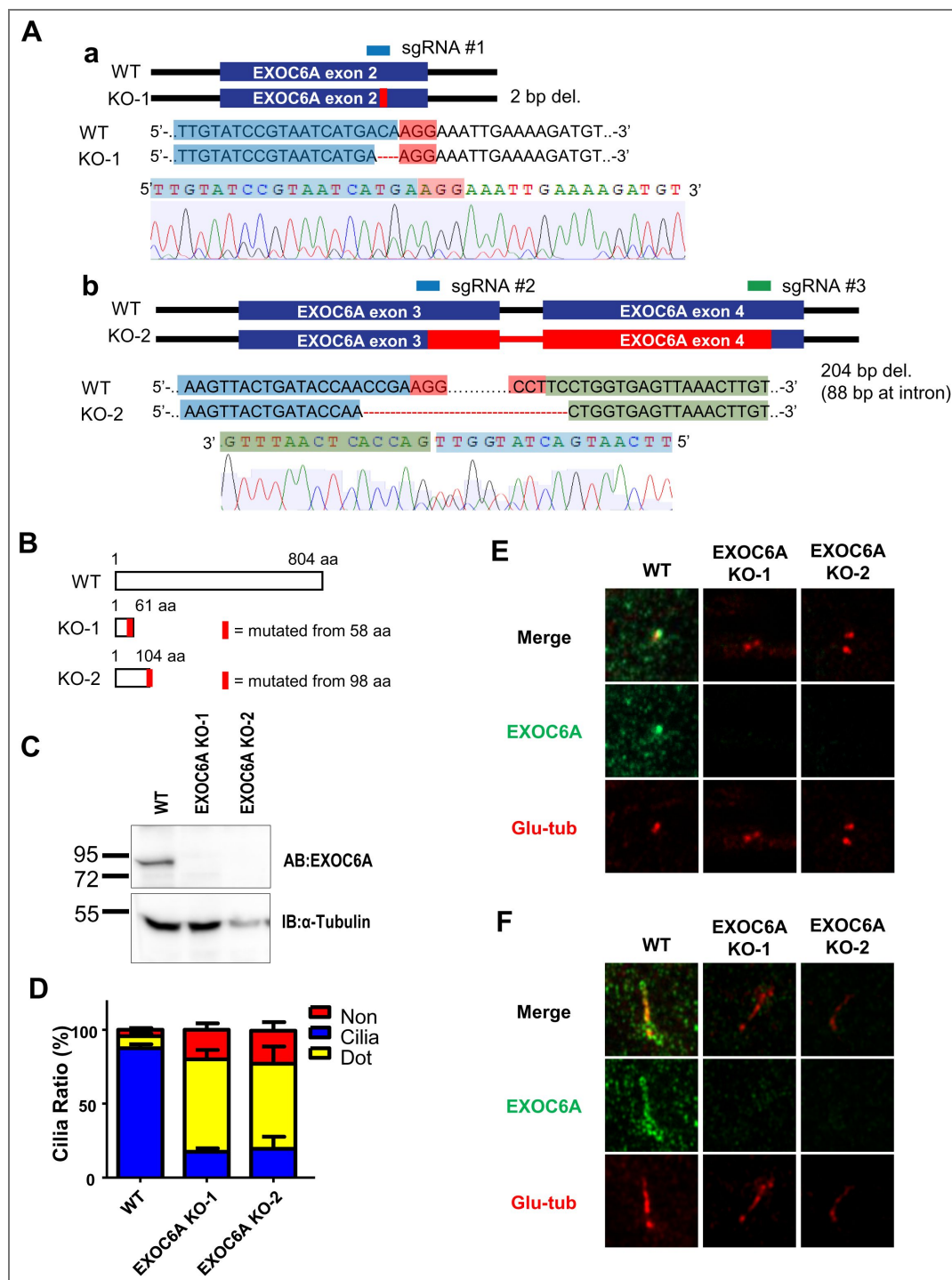
Supplementary figures



Supplementary Figure 1. Correlation of spatial localization of EXOC6A and Myo-Va during the early stages of ciliogenesis. (A) RPE1-based mCherry-Myo-Va-GTD-inducible cells were treated with DOX for 24 h and serum starved for 0, 15, and 30 min. Fluorescence intensities of EXOC6A (green) or Myo-Va-GTD (red) signals within a 2 μ m radius surrounding the Glu-tub-labeled centrioles (white) were quantified and are shown in B (EXOC6A) and C (Myo-Va-GTD). Correlation R (Pearson correlation coefficient) between EXOC6A and Myo-Va is shown in D. Correlation R was analyzed using Zeiss Zen blue software. Error bars in B–D represent mean $\pm$ s.d. from at least 3 independent experiments with 100 randomly selected cells. *P*-value was determined with one-way ANOVA. $P < 0.05$ was considered statistically significant. NS, not significant. Scale bars are 1 μ m.

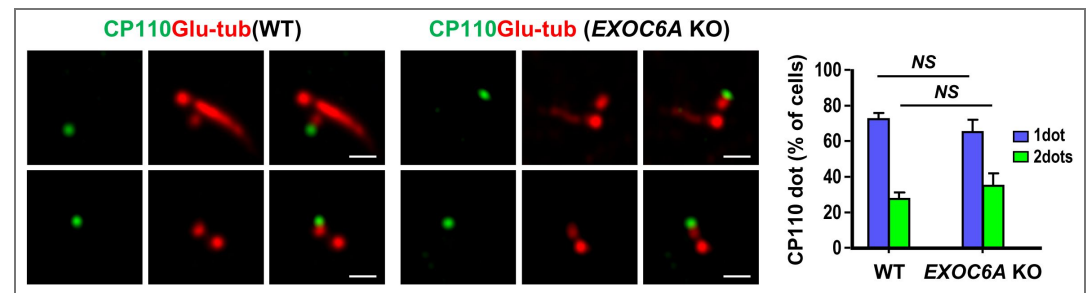


Supplementary Figure 2. Live-cell imaging of GFP-EXOC6A co-localized with Myo-Va-GTD at the ciliary membrane (A) RPE1-based inducible cells expressing GFP-EXOC6A and mCherry-MyoVa-GTD were treated with Dox for 24 h, serum starved for 30 min, and SiR-tubulin was added to label centriole and axoneme. (B) Single channel image (green box in A) captured at the starting time point (0 sec). (C) Single channel image (red box in A) captured at frame 21 (197.577 sec). Scale bars are 1 μ m.

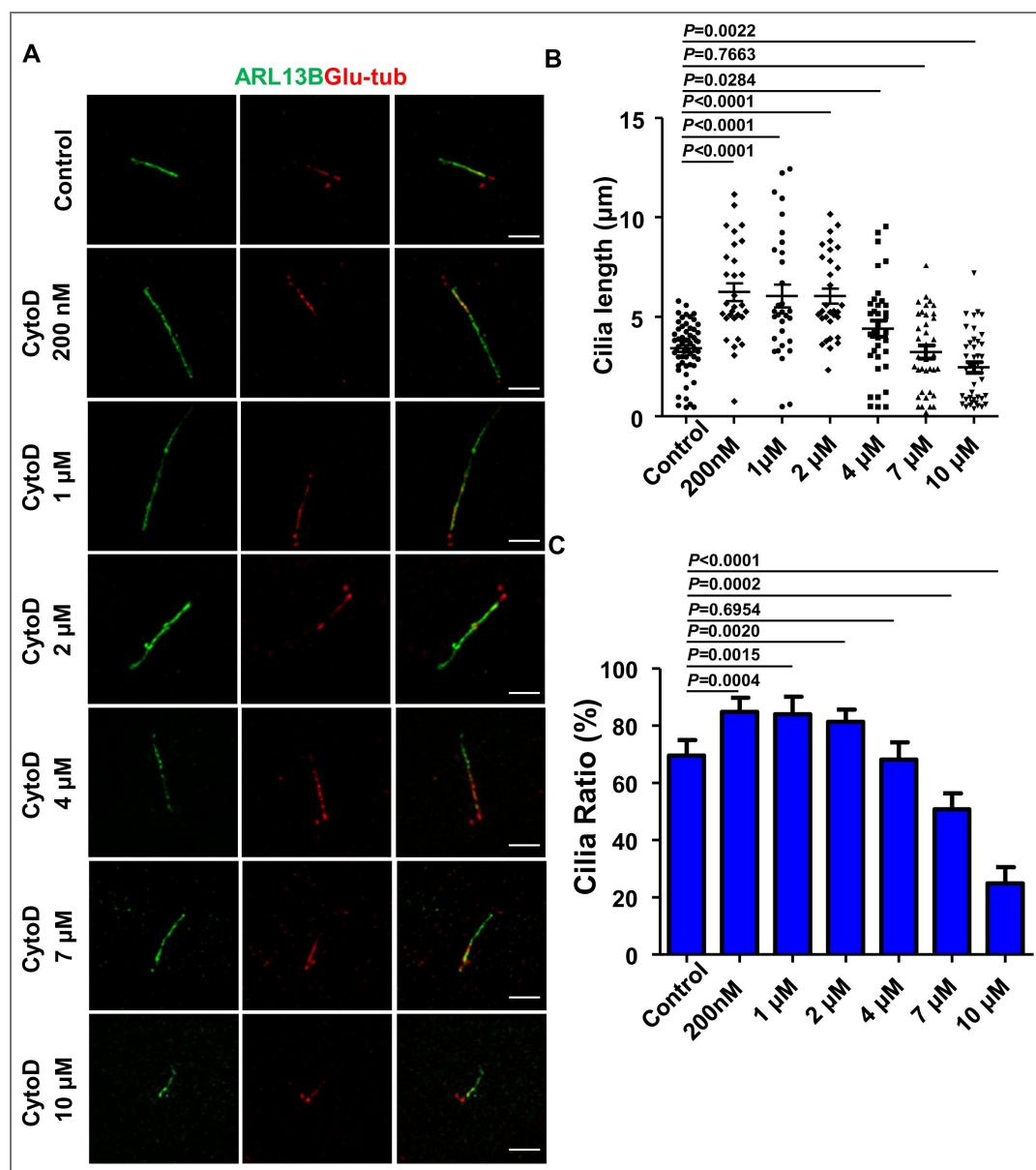


Supplementary Figure 3. Construction and Characterization of EXOC6A Knockout Cell Lines (A) Schematic illustration of the CRISPR-Cas9 gene-editing strategies used to construct *EXOC6A*-KO RPE1 cells. (a) Construction of the KO-1 clone: using a single sgRNA1 targeting exon 2 of the *EXOC6A* gene. Gene sequence analysis confirms a 2 bp deletion (del), resulting in a frameshift mutation. (b) Construction of the KO-2 clone: using a dual-sgRNA strategy (sgRNA 2 targeting exon 3 and sgRNA 3 targeting exon 4). Gene sequence analysis confirms a large 204 bp deletion spanning the intron between exon 3 and 4 (including 88 bp at the intron). (B) Schematic diagram of the predicted protein products. Frameshift mutations in KO-1 and KO-2 were expected to lead to premature termination compared to the full-length Wild-Type (WT) protein (804 aa), resulting in truncated non-functional peptides of 61 aa and 104 aa, respectively. (C) Immunoblot validation of knockout efficiency. Western blot analysis showed complete loss of *EXOC6A* protein in both KO-1 and KO-2 clones compared to WT RPE1 cells. (D) Quantitative analysis of cilia ratio in WT and *EXOC6A* KO clones. WT and KO Cells were treated with serum starvation for 24 h. Ciliary structures were visualized by ARL13B staining and categorized as cilia (blue), dot

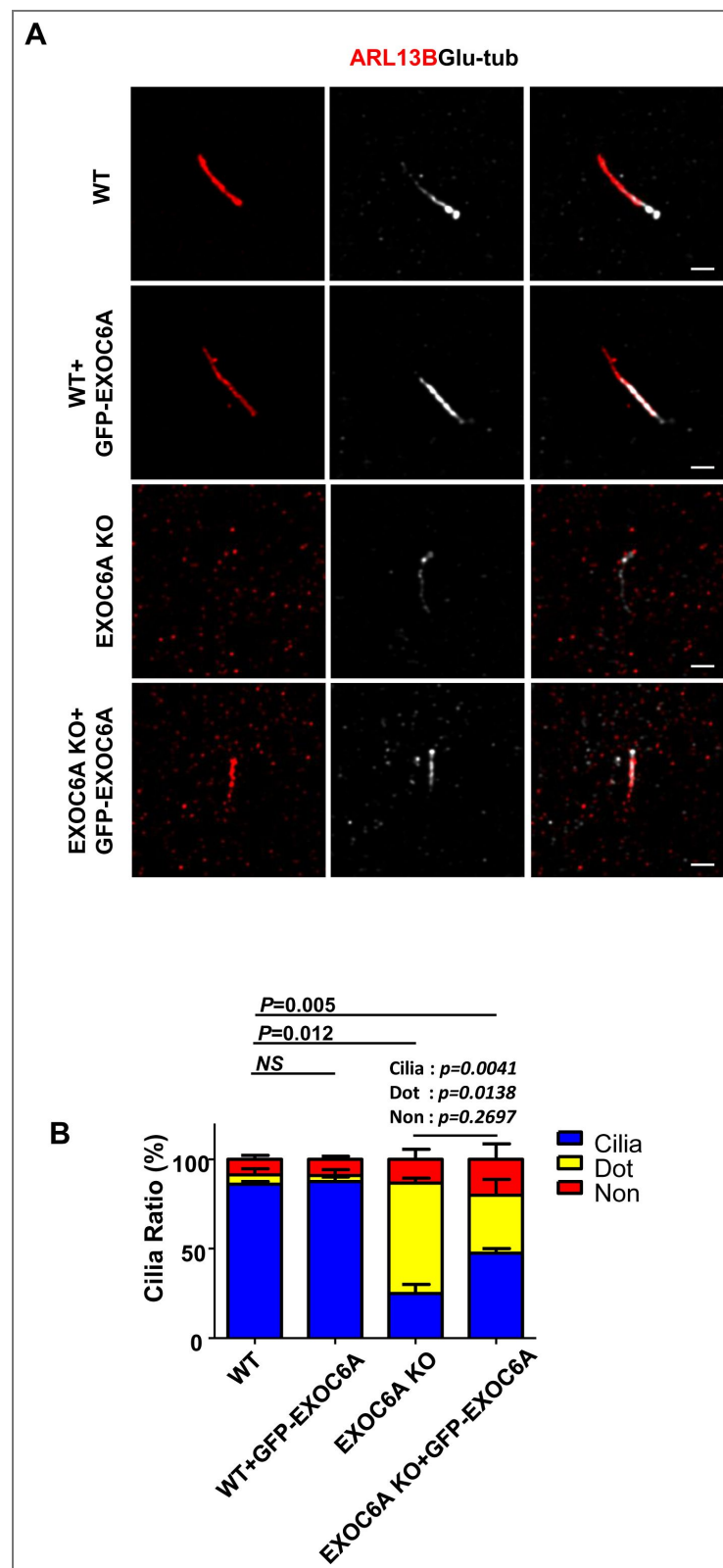
(yellow), or no structure (Non, red). Notably, both independent KO clones exhibited similar phenotypes, predominantly arresting at the "Dot" stage. Error bars represent the mean $\pm$ s.d. from three independent experiments (n=100 cells per experiment). (E and F) Immunofluorescence assays confirmed antibody specificity and the absence of protein expression in *EXOC6A* knockout clones. Wild-type cells and two independent *EXOC6A* knockout clones (KO-1 and KO-2) were serum-starved for 2h (E, for the CV phase) or 24 h (F, for the ciliary phase) to induce ciliogenesis. WT cells showed clear *EXOC6A* localization at the CV or ciliary sheath, while no specific green signal was detected in either KO clone.



Supplementary Figure 4. *EXOC6A* deletion does not interfere with the removal of CP110 from the mother centriole. WT and *EXOC6A* KO cells were fixed 24 h after serum starvation and analyzed via fluorescence confocal microscopy using the indicated antibodies. (B) Percentages of cells with one or two CP110 dots are shown. Error bars represent the mean $\pm$ s.d. from at least 3 independent experiments with 100 cells per experiment. *P*-value was determined with two-tailed Student's *t*-test. NS, not significant. Scale bars are 1 μ m.



Supplementary Figure 5. Low doses of Cytochalasin D (CytoD) promote cilia elongation, whereas higher concentrations (greater than 4 μM) inhibit ciliogenesis. (A) RPE1 cells were treated with different concentrations of Cytochalasin D (CytoD), including control (0.1% DMSO), 200 nM, 1 μM, 2 μM, 4 μM, 7 μM, and 10 μM. Quantification of ciliary length labeled with ARL13B antibody (B) and cilia ratio (C) are shown. P-values were determined with a two-tailed Student's *t*-test. $P < 0.05$ was considered statistically significant. Error bars represent the mean $\pm$ s.d. of 100 randomly selected cells from at least 3 independent experiments. Scale bars are 2 μm.



Supplementary Figure 6. Rescue of ciliogenesis defects in EXOC6A knockout cells by GFP-EXOC6A re-expression. (A) Representative immunofluorescence images of RPE1 cells (WT, WT+GFP-EXOC6A, EXOC6A KO, and EXOC6A KO+GFP-EXOC6A) stained for ARL13B (red) and Glu-tubulin (white). (B) Quantitative analysis of ARL13B-labeled cilia phenotypes. Data are presented as the mean $\pm$ s.d. of 100 randomly selected cells from at least three independent experiments. P -value was determined with two-tailed Student's t -test. $P < 0.05$ was considered statistically significant. NS, not significant. Scale bars are 1 μ m.

Uncropped images of Western blots shown in the main figures

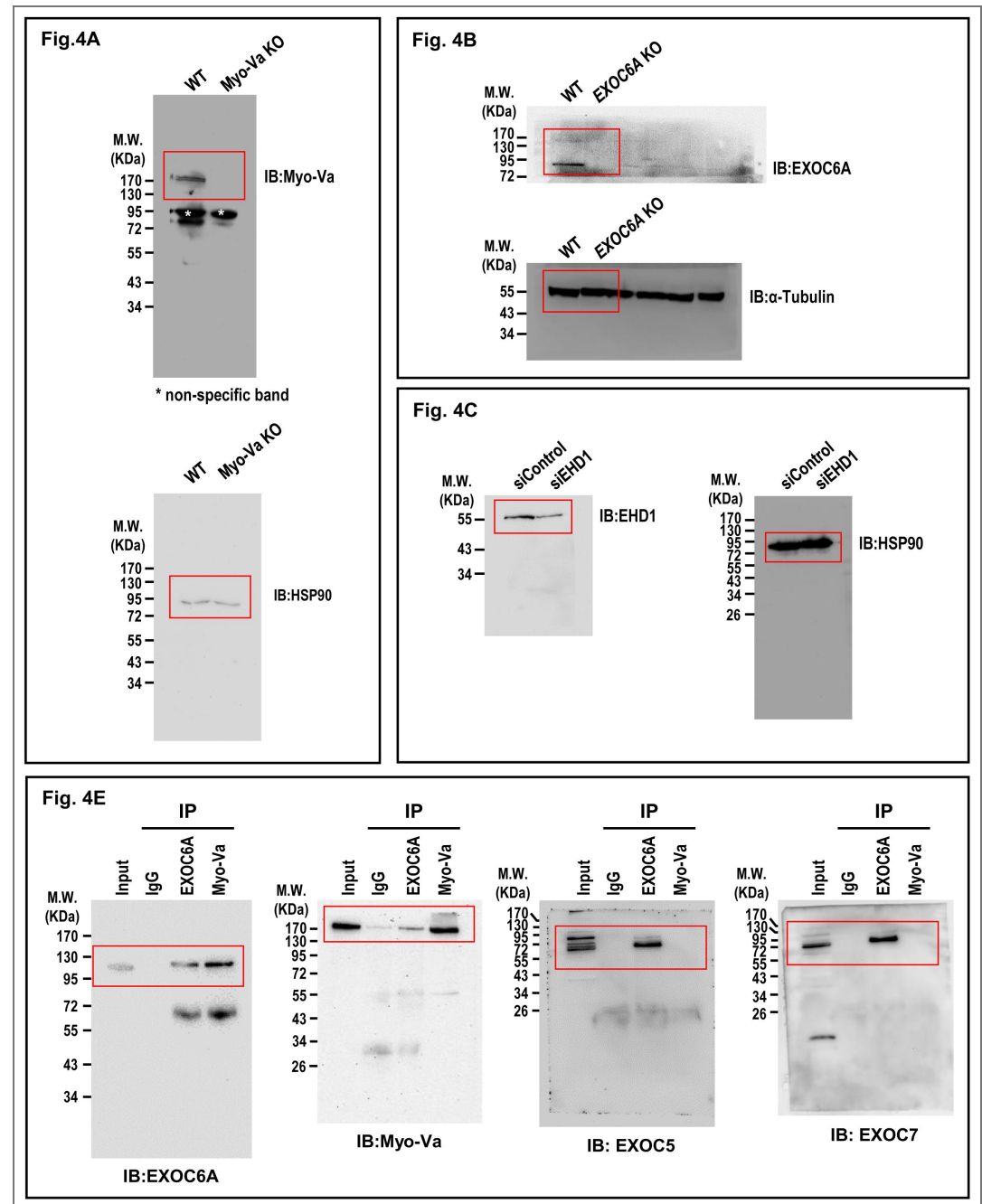


Fig. 5B

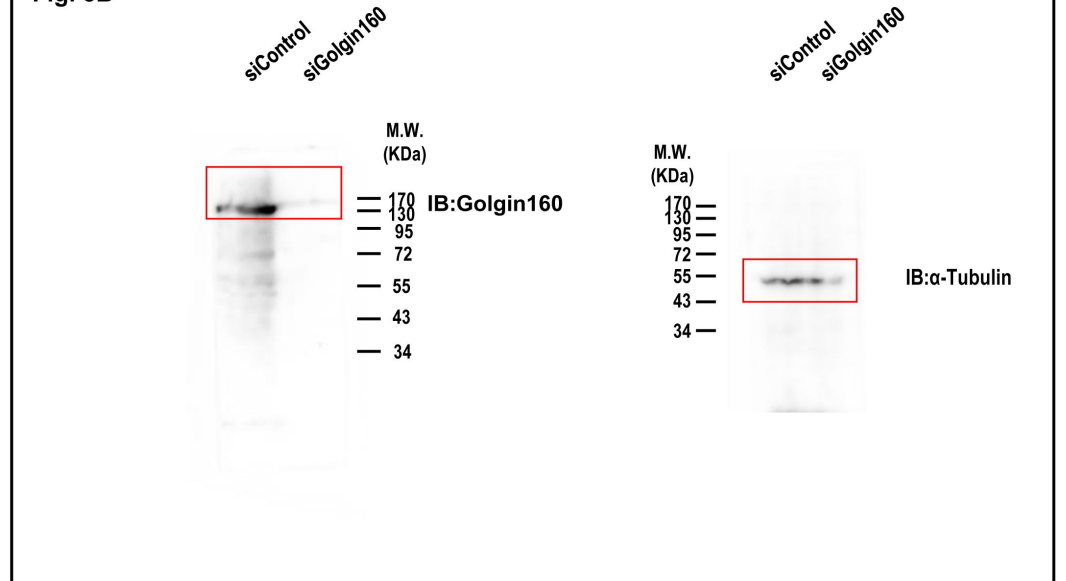
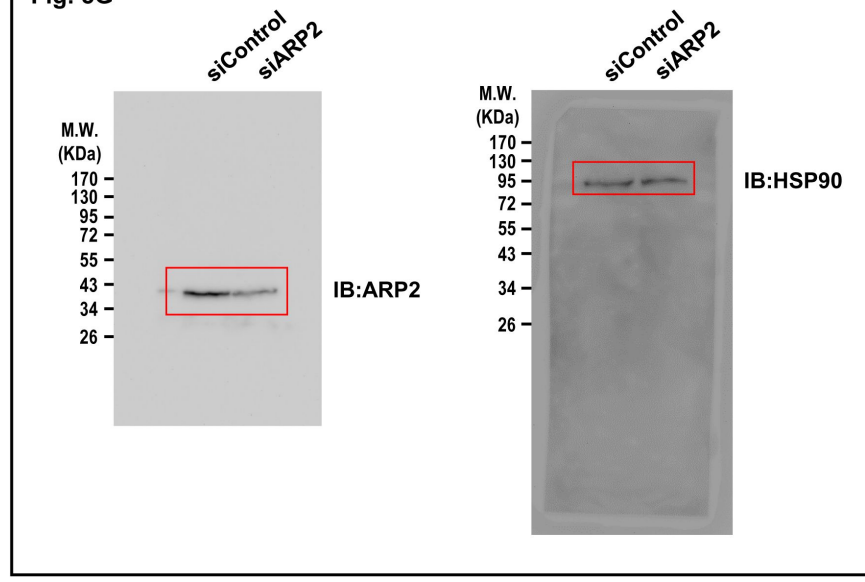
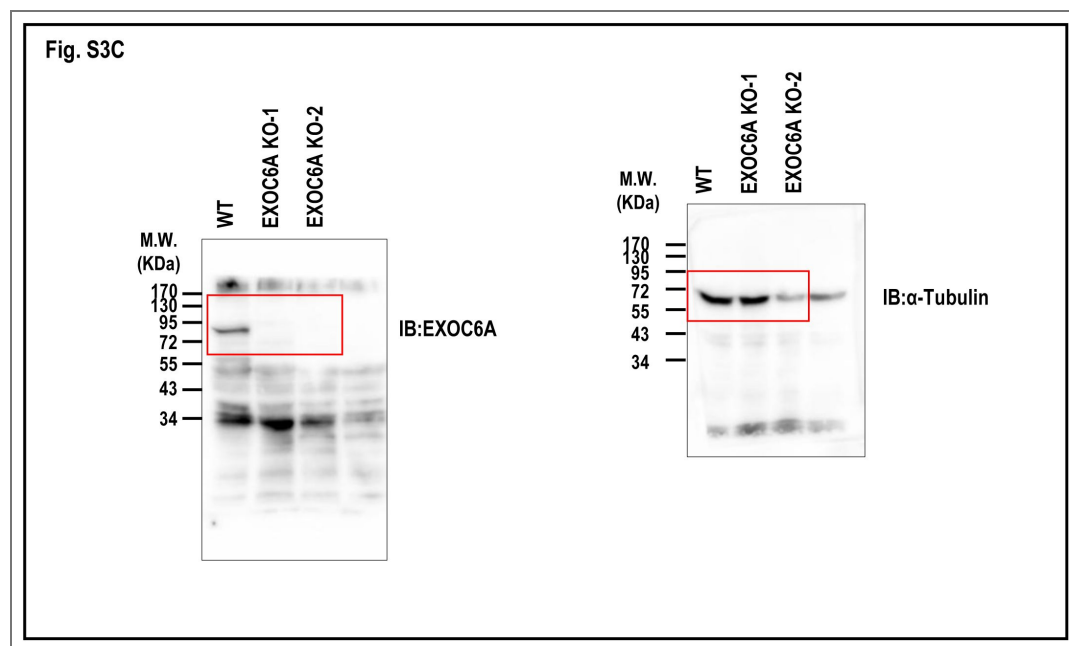


Fig. 5G





Data availability

All data generated or analyzed during this study are included in the manuscript and supporting files

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Additional information

Author contributions

Conceptualization: TLL and TKT; Investigation: TLL and CTW; Methodology: TLL, CTW, and TKT; Resource: TLL and TKT; Original Draft: TLL and TKT; Writing: TLL and TKT; Review & Editing: TLL, CTW, and TKT.

Author ORCID iDs

Tang K Tang: <https://orcid.org/0000-0002-3660-0695>

Additional files

Video 1. [Related to Fig. 3A.](#) RPE1-based inducible cells expressing GFP-EXOC6A and mCherry-Myo-Va-GTD were treated with Dox for 24 h and then serum starved for 30 min. Live cell images were taken using an LSM880 Carl Zeiss confocal system under a controlled CO₂ (5%) and temperature (37 °C). Photobleaching was performed on the CVs. The first time point of photobleaching the CVs was at 2 min, following which the CVs gradually recovered.

Video 2. [Related to Fig. 3B.](#) RPE1-based inducible cells expressing GFP-EXOC6A and mCherry-Myo-Va-GTD were treated with Dox for 24 h and then serum starved for 30 min. Live cell images were taken using an LSM880 Carl Zeiss confocal system under a controlled CO₂ (5%) and

temperature (37 °C). Photobleaching was performed on the ciliary membrane. The first photobleaching time point for the ciliary membrane was at 2 min, after which the ciliary membrane gradually recovered.

Video 3. [Related to Fig. S2.](#) RPE1-based inducible cells expressing GFP-EXOC6A and mCherry-MyoVa-GTD were treated with Dox for 24 h, then serum starved for 30 min, and SiR-tubulin was added to label centriole and axoneme. Live cell images were acquired using an Elyra 7 SIM Carl Zeiss confocal system under a controlled CO₂ (5%) and temperature (37 °C).

Video 4. [Related to Fig. 3C.](#) RPE1-based inducible cells expressing GFP-EXOC6A were treated with Dox for 24 h and then serum starved for 30 min. Live cell images were taken using an Elyra 7 SIM Carl Zeiss confocal system under a controlled CO₂ (5%) and temperature (37 °C).

Video 5. [Related to Fig. 3D.](#) RPE1-based inducible cells expressing GFP-EXOC6A were treated with Dox for 24 h and then serum starved for 30 min. Live cell images were taken using an Elyra 7 SIM Carl Zeiss confocal system under a controlled CO₂ (5%) and temperature (37 °C).

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Peer reviews

Reviewer #2 (Public review):

Summary:

The molecular mechanisms underlying ciliogenesis are not well understood. Previously, work from the same group (Wu et al., 2018) identified myosin-Va as an important protein in transporting preciliary vesicles to the mother vesicles, allowing for initiation of ciliogenesis. The exocyst complex has previously been implicated in ciliogenesis and protein trafficking to cilia. Here, Lin et al. investigate the role of exocyst complex protein EXOC6A in cilia formation. The authors find that EXOC6A localizes to preciliary vesicles, ciliary vesicles, and the ciliary sheath. EXOC6A colocalizes with Myo-Va in the ciliary vesicle and the ciliary sheath, and both proteins are removed from fully assembled cilia. EXOC6A is not required for Myo-Va localization, but Myo-Va and EHD1 are required for EXOC6A to localize in ciliary vesicles. The authors propose that EXOC6A vesicles continually remodel the cilium: FRAP analysis demonstrates that EXOC6A is a dynamic protein, and live imaging shows that EXOC6A fuses with and buds off from the ciliary membrane. Loss of EXOC6A reduces, but does not eliminate, the number of cilia formed in cells. Any cilia that are still present are structurally abnormal, with either bent morphologies or transition zone defects. Overall, the analyses and imaging are well done, and the conclusions are well supported by the data. The work will be of interest to cell biologists, especially those interested in centrosomes and cilia.

Strengths:

The TEM micrographs are of excellent quality. The quality of the imaging overall is very good, especially considering that these are dynamic processes occurring in a small region of the cell. The data analysis is well done and the quantifications are very helpful. The manuscript is well-written and the final figure is especially helpful in understanding the model.

The manuscript has greatly improved after revision. In particular, testing GPR161 and BBS9 localization is helpful evidence to demonstrate that transition zone function is disrupted when EXOC6A is lost. The generation of a second knockout clone and tests of antibody specificity are also great additions.

Weaknesses:

None

<https://doi.org/10.7554/eLife.108271.2.sa2>

Reviewer #3 (Public review):

Summary:

Lin et al report on the dynamic localization of EXOC6A and Myo-Va at pre-ciliary vesicles, ciliary vesicles, and ciliary sheath membrane during ciliogenesis using three-dimensional structured illumination microscopy and ultrastructure expansion microscopy. The authors

further confirm the interaction of EXOC6A and Myo-Va by co-immunoprecipitation experiments and demonstrated the requirement of EHD1 for the EXOC6A-labeled ciliary vesicles formation. Additional experiments using gene-silencing by siRNA and pharmacological tools identified the involvement of dynein-, microtubule-, and actin in the transport mechanism of EXOC6A-labeled vesicles to the centriole, as they have previously reported for Myo-Va. Notably, loss of EXOC6A severely disrupts ciliogenesis, with the majority of cells becoming arrested at the ciliary vesicle (CV) stage, highlighting the involvement of EXOC6A at later stages of ciliogenesis. As the authors observe dynamic EXOC6A-positive vesicle release and fusion with the ciliary sheath, this suggests a role in membrane and potentially membrane protein delivery to the growing cilium past the ciliary vesicle stage. While CEP290 localization at the forming cilium appears normal the recruitment of other transition zone components, exemplified by several MKS and NPHP module components, was also impaired in EXOC6A-deficient cells.

Strengths:

- By applying different microscopy approaches, the study provides deeper insight into the spatial and temporal localization of EXOC6A and Myo-Va during ciliogenesis.
- The combination of complementary siRNA and pharmacological tools targeting different components strengthens the conclusions.
- This study reveals a new function of EXOC6A in delivering membrane and membrane proteins during ciliogenesis, both to the ciliary vesicle as well as to the ciliary sheath.
- The overall data quality is high. The investigation of EXOC6A at different time points during ciliogenesis is well schematized and explained.
- The authors confirmed central antibody reagents used in this study and validated key experiments by using two independent knockout clones (for which sequencing information was provided).

Weaknesses:

- The precise molecular function of EXOC6A remains open, as the presented data suggests no involvement of other exocyst components.

Taken together, the authors achieved their goal to elucidate the role of EXOC6A in ciliogenesis, demonstrating its involvement in vesicle trafficking and membrane remodeling in both early and late stages of ciliogenesis. Their findings are supported by experimental evidence. This work is likely to have an impact on the field by expanding our understanding of the molecular machinery underlying cilia biogenesis, particularly the coordination between exocyst components and cytoskeletal transport systems. The methods and data presented offer valuable tools for dissecting vesicle dynamics and cilium formation, providing a foundation for future research into ciliary dysfunction and related diseases. By connecting vesicle trafficking to structural maturation of an organelle, the study adds important context to the broader description of cellular architecture and organelle biogenesis.

Comments on revisions:

We very much appreciate the extra work you put into improving your manuscript and want to congratulate you on your important discovery. We encourage you to keep up the good work!

<https://doi.org/10.7554/eLife.108271.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

Summary:

The study by Lin et al. studies the role of EXOC6A in ciliogenesis and its relationship with the interactor myosin-Va using a range of approaches based on the RPE1 cell line model. They establish its spatio-temporal organization at centrioles, the forming ciliary vesicle and ciliary sheath using ExM, various super-resolution techniques, and EM, including correlative light and electron microscopy. They also perform live imaging analyses and functional studies using RNAi and knockout. They establish a role of EXOC6A together with myosin-Va in Golgi-derived, microtubule- and actin-based vesicle trafficking to and from the ciliary vesicle and sheath membranes. Defects in these functions impair robust ciliary shaft and axoneme formation due to defective transition zone assembly.

Strengths:

The study provides very high-quality data that support the conclusions. In particular, the imaging data is compelling. It also integrates all findings in a model that shows how EXOC6A participates in multiple stages of ciliogenesis and how it cooperates with other factors.

Weaknesses:

The precise role of EXOC6A remains somewhat unclear. While it is described as a component of the exocyst, the authors do not address its molecular functions and whether it indeed works as part of the exocyst complex during ciliogenesis.

We sincerely thank Reviewer 1 for the thoughtful evaluation of our manuscript and the constructive comments provided. We are especially grateful for the recognition of the quality and significance of our imaging data and the comprehensive model we propose regarding EXOC6A's role in ciliogenesis. We did not address the function of other components of the exocyst complex during ciliogenesis. However, in our biochemical analyses, Myosin-Va specifically co-immunoprecipitated with EXOC6A but not with other exocyst subunits tested (EXOC5 and EXOC7) (Fig. 4E) indicating a selective interaction between EXOC6A and the Myo-Va transport machinery.

Reviewer #2 (Public review):

Summary:

The molecular mechanisms underlying ciliogenesis are not well understood. Previously, work from the same group (Wu et al., 2018) identified myosin-Va as an important protein in transporting preciliary vesicles to the mother vesicles, allowing for initiation of ciliogenesis. The exocyst complex has previously been implicated in ciliogenesis and protein trafficking to cilia. Here, Lin et al. investigate the role of exocyst complex protein EXOC6A in cilia formation. The authors find that EXOC6A localizes to preciliary vesicles, ciliary vesicles, and the ciliary sheath. EXOC6A colocalizes with Myo-Va in the ciliary vesicle and the ciliary sheath, and both proteins are removed from fully assembled cilia. EXOC6A is not required for Myo-Va localization, but Myo-Va and EHD1 are required for EXOC6A to localize in ciliary vesicles. The authors propose that EXOC6A vesicles continually remodel the cilium: FRAP analysis demonstrates that EXOC6A is a dynamic

protein, and live imaging shows that EXOC6A fuses with and buds off from the ciliary membrane. Loss of EXOC6A reduces, but does not eliminate, the number of cilia formed in cells. Any cilia that are still present are structurally abnormal, with either bent morphologies or the absence of some transition zone proteins. Overall, the analyses and imaging are well done, and the conclusions are well supported by the data. The work will be of interest to cell biologists, especially those interested in centrosomes and cilia.

Strengths:

The TEM micrographs are of excellent quality. The quality of the imaging overall is very good, especially considering that these are dynamic processes occurring in a small region of the cell. The data analysis is well done and the quantifications are very helpful. The manuscript is well-written and the final figure is especially helpful in understanding the model.

Weaknesses:

Additional information about the functional and mechanistic roles of EXOC6A would improve the manuscript greatly.

We sincerely thank Reviewer 2 for the thoughtful and encouraging evaluation of our work. We are grateful for the recognition of the strengths of our study, including the quality of the TEM micrographs, the rigor of our imaging and data analysis, and the clarity of our manuscript and proposed model.

We have expanded our analyses in the revised manuscript to better define EXOC6A's contribution to ciliary function. Specifically, we examined the trafficking of two critical ciliary membrane-associated proteins: GPR161, a G-protein-coupled receptor involved in Sonic hedgehog (Shh) signaling, and BBS9, a core component of the BBSome complex essential for ciliary membrane protein transport. Our new data (Fig. 7C) show that both GPR161 and BBS9 fail to localize to the cilium in EXOC6A knockout cells, in contrast to wild-type controls where their ciliary localization is robust. This new evidence significantly strengthens the understanding of EXOC6A's role.

Reviewer #3 (Public review):

Summary:

Lin et al report on the dynamic localization of EXOC6A and Myo-Va at pre-ciliary vesicles, ciliary vesicles, and ciliary sheath membrane during ciliogenesis using three-dimensional structured illumination microscopy and ultrastructure expansion microscopy. The authors further confirm the interaction of EXOC6A and Myo-Va by co-immunoprecipitation experiments and demonstrated the requirement of EHD1 for the EXOC6A-labeled ciliary vesicles formation. Additional experiments using gene-silencing by siRNA and pharmacological tools identified the involvement of dynein-, microtubule-, and actin in the transport mechanism of EXOC6A-labeled vesicles to the centriole, as they have previously reported for Myo-Va. Notably, loss of EXOC6A severely disrupts ciliogenesis, with the majority of cells becoming arrested at the ciliary vesicle (CV) stage, highlighting the involvement of EXOC6A at later stages of ciliogenesis. As the authors observe dynamic EXOC6A-positive vesicle release and fusion with the ciliary sheath, this suggests a role in membrane and potentially membrane protein delivery to the growing cilium past the ciliary vesicle stage. While CEP290 localization at the forming cilium appears normal, the recruitment of other transition zone components, exemplified by several MKS and NPHP module components, was also impaired in EXOC6A-deficient cells.

Strengths:

(1) By applying different microscopy approaches, the study provides deeper insight into the spatial and temporal localization of EXOC6A and Myo-Va during ciliogenesis.

(2) The combination of complementary siRNA and pharmacological tools targeting different components strengthens the conclusions.

(3) This study reveals a new function of EXOC6A in delivering membrane and membrane proteins during ciliogenesis, both to the ciliary vesicle as well as to the ciliary sheath.

(4) The overall data quality is high. The investigation of EXOC6A at different time points during ciliogenesis is well schematized and explained.

Weaknesses:

(1) Since many conclusions are based on EXOC6A immunostaining, it would strengthen the study to validate antibody specificity by demonstrating the absence of staining in EXOC6A-deficient cells.

(2) While the authors generated an EXOC6A-deficient cell line, off-target effects can be clone-specific. Validating key experiments in a second independent knockout clone or rescuing the phenotype of the existing clone by re-expressing EXOC6A would ensure that the observed phenotypes are due to EXOC6A loss rather than unintended off-target effects.

(3) Some experimental details are lacking from the materials and methods section. No information on how the co-immunoprecipitation experiments have been performed can be found. The concentrations of pharmacological agents should be provided to allow proper interpretation of the results, as higher or lower doses can produce nonspecific effects. For example, the concentrations of ciliobrevin and nocodazole used to treat RPE1 cells are not specified and should be included. More precise settings for the FRAP experiments would help others reproduce the presented data. Some details for the siRNA-based knockdowns, such as incubation times, can only be found in the figure legends.

Taken together, the authors achieved their goal of elucidating the role of EXOC6A in ciliogenesis, demonstrating its involvement in vesicle trafficking and membrane remodeling in both early and late stages of ciliogenesis. Their findings are supported by experimental evidence. This work is likely to have an impact on the field by expanding our understanding of the molecular machinery underlying cilia biogenesis, particularly the coordination between the exocyst complex and cytoskeletal transport systems. The methods and data presented offer valuable tools for dissecting vesicle dynamics and cilium formation, providing a foundation for future research into ciliary dysfunction and related diseases. By connecting vesicle trafficking to structural maturation of an organelle, the study adds important context to the broader description of cellular architecture and organelle biogenesis.

We sincerely thank Reviewer 3 for the thorough and thoughtful assessment of our manuscript. We greatly appreciate the recognition of the strengths of our study, including the use of advanced microscopy techniques, complementary functional tools, and the conceptual contributions regarding EXOC6A's role in vesicle trafficking and membrane remodeling during ciliogenesis.

Below, we detail how we have addressed the specific suggestions for improvement:

(1) Validation of EXOC6A Immunostaining Specificity

To directly address the reviewer's concern regarding antibody specificity, we have included new control immunofluorescence panels in Figure S3E-F, which show a complete loss of EXOC6A signal in two independent knockout (KO) clones. These data confirm the specificity of the EXOC6A antibody used throughout the study and reinforce the accuracy of our localization analyses at different stages of ciliogenesis.

(2) Addressing Potential Clone-Specific or Off-Target Effects

To ensure that the observed phenotypes are attributable to EXOC6A loss and not due to off-target effects, we performed parallel analyses using two independent KO clones, all of which exhibited identical defects in ciliogenesis, including arrest at the ciliary vesicle stage and impaired cilia assembly (Fig. S3C-D).

In addition, we conducted rescue experiments by re-expressing EXOC6A in the KO background, which effectively restored ciliogenesis. Quantitative analysis of the rescue data has been added to the revised manuscript (Figure S6B), providing further support that the observed phenotype is specifically due to EXOC6A deficiency.

(3) Expanded Methodological Details

- A detailed protocol for co-immunoprecipitation experiments, including lysis conditions, antibody concentrations, and washing steps.
- The precise concentrations and treatment durations for all pharmacological agents used, including ciliobrevin and nocodazole.
- Comprehensive details on the siRNA-mediated knockdowns, including oligonucleotide sequences, transfection reagents, and incubation durations.

Recommendations for the authors:

Reviewing Editor Comments:

After further consultation, all 3 reviewers agreed that this is an important study with highquality data, in particular the imaging data. They also considered most of the evidence convincing, but overall they termed it "solid" for two main reasons: first, they would have liked to see a validation of the EXOC6A antibody specificity, and second, they suggest that you demonstrate for at least key experiments the phenotypes with a second KO clone, to exclude clonal effects. In principle, rescue would be suited to address this, but the issue here is that the presented rescue is not very robust.

We sincerely thank the Editor and all reviewers for their constructive and thoughtful evaluation of our manuscript. We are especially grateful for the recognition of the highquality imaging data, the experimental rigor, and the significance of our findings to the field of ciliogenesis.

We fully acknowledge the two principal concerns raised during further consultation: (1) the need for validation of EXOC6A antibody specificity, and (2) the importance of confirming the phenotypes in an independent knockout clone to exclude clonal artifacts. We have taken both of these points seriously and have now addressed them through additional experiments and analyses, as detailed below:

(1) Validation Using Independent Knockout Clones

To rigorously validate antibody specificity and eliminate the possibility of clonal variation, we have characterized a second independent EXOC6A knockout (KO) clone. We confirmed complete loss of EXOC6A expression in both clones using three orthogonal approaches: genotyping, immunoblotting, and immunofluorescence (Fig. S3). Both KO clones exhibit

indistinguishable phenotypes, including arrest at the ciliary vesicle stage and impaired cilia formation (Fig. S3D).

(2) Rescue Phenotype Validation with Statistical Significance

In response to concerns about the robustness of the rescue, we have now included statistical analysis of the rescue experiments. A two-tailed Student's t-test comparing ciliogenesis between the EXOC6A KO and rescue (GFP-EXOC6A re-expression) conditions shows a statistically significant improvement ($p = 0.0041$) (Fig. S6B). While we acknowledge that the rescue is partial—likely due to limitations of overexpression systems—the statistically significant recovery provides strong genetic evidence that the phenotypes are specific and reversible. These data are now included in the revised Figure S6.

(3) Functional Consequences of EXOC6A Loss on Ciliary Membrane Protein Trafficking

To further strengthen the mechanistic conclusions, we expanded our study to include the trafficking of two functional ciliary membrane proteins. We show that in EXOC6A KO cells, both BBS9 (a component of the BBSome complex) and GPR161 (a GPCR involved in Shh signaling) fail to enter the cilium. These results suggest that EXOC6A is required not only for early structural events in ciliogenesis, but also for establishing a competent transition zone, critical for ciliary membrane protein recruitment. These findings are detailed in the revised Figure 7C and corresponding Results.

We believe that these additional experiments and clarifications directly address the concerns and significantly strengthen the robustness and impact of our study.

The reviewers also made additional suggestions regarding functional and mechanistic insights that would strengthen the manuscript even further.

Reviewer #1 (Recommendations for the authors):

(1) The authors should include control IF panels for the specificity of the EXOC6A stainings at the various ciliogenesis stages using the KO cell line.

We thank the reviewer for this important suggestion. We have now included the requested immunofluorescence (IF) control panels to validate the specificity of the EXOC6A antibody. As shown in the newly added Figure S3, EXOC6A immunofluorescence signal is completely absent in EXOC6A knockout (KO) cells at CV (Fig. S3E) and cilia membrane (Fig. S3F) stages, whereas robust and stage-specific signals are observed in wild-type cells. These results confirm the specificity of the endogenous EXOC6A staining used throughout the study and validate the spatiotemporal localization patterns reported in the main figures.

(2) It would be informative to compare EXOC6A KO and RNAi to determine whether the only partially impaired ciliogenesis phenotype may be a consequence of cellular adaptation.

We appreciate the reviewer's concern regarding potential cellular adaptation or clonespecific effects. To address this, we examined the ciliogenesis phenotype in two independent EXOC6A KO clones generated using distinct sgRNA targeting strategies. As shown in Figure S3, two independent KO clones displayed a highly consistent phenotype characterized by a pronounced arrest at the ciliary vesicle (CV) stage and a significant reduction in mature cilium formation.

The reproducibility of this phenotype across multiple independently derived clones strongly argues against clonal variability or long-term adaptive compensation as the underlying cause. Instead, these results support the conclusion that the observed ciliogenesis defects are a direct and specific consequence of EXOC6A loss.

(3) It remains unclear whether EXOC6A's function in ciliogenesis is part of the exocyst complex. This is currently implied by the context in which it is introduced and discussed, although the authors avoid any direct statement about this. Do the authors observe similar phenotypes by knocking down any other exocyst subunit? In any case, this issue should be discussed.

We thank the reviewer for raising this conceptual point. This study did not explore the functions of other components of the exocytosis complex during ciliogenesis, which warrants further investigation in the future. However, in our biochemical analyses, Myosin -Va specifically co-immunoprecipitated with EXOC6A but not with other exocyst subunits tested (EXOC5 and EXOC7) (Fig. 4E) indicating a selective interaction between EXOC6A and the Myo-Va transport machinery.

Reviewer #2 (Recommendations for the authors):

To clarify the roles of EXOC6A in ciliogenesis, I suggest the following:

(1) Myo-Va is involved in both the intracellular and extracellular ciliogenesis pathways. The authors show that EXOC6A has a role in the intracellular ciliogenesis pathway. Does it also participate in the extracellular pathway?

We thank the reviewer for this insightful question. Given that Myo-Va functions in both intracellular and extracellular ciliogenesis pathways, it is indeed plausible that EXOC6A may also participate in the extracellular pathway. However, the current study was specifically focused on elucidating the molecular mechanisms of intracellular ciliogenesis using RPE1 cells, which exclusively undergo this pathway. Assessing EXOC6A's role in the extracellular pathway would require the use of specialized models (e.g., polarized epithelial cells such as MDCK or IMCD3), which fall beyond the scope of this manuscript.

(2) In the live imaging movies (Fig 3C, 3D, supp movie 4 and 5), the authors observe tubular structures and puncta with EXOC6A and conclude that these are dynamic vesicles/membranes. While the movies are suggestive of membrane-like behavior, it would be helpful to show that these puncta and tubules have membrane, perhaps by staining with a membrane dye.

We appreciate the reviewer's suggestion to validate the membrane identity of EXOC6A-positive structures. While we did not perform membrane dye staining in the current study, we agree this approach would provide additional confirmation. Nevertheless, the dynamic behaviors observed in our live-cell imaging—including membrane-like tubulation, fusion, and fission—strongly support the interpretation that EXOC6A puncta and tubules

(3) It is unclear how the EXOC6A tubules and vesicles are delivered, and the extent to which MyoVa plays a role. The authors co-label EXOC6A and MyoVa in Supp Fig 2, but EXOC6A dynamics seem very different here, as compared to Fig 3D - there are fewer tubules and puncta and less movement of either tubules or puncta between time points. Does expression of MyoVa decrease EXOC6A membrane dynamics? Or is it required for EXOC6A membrane dynamics?

We thank the reviewer for this observation. The apparent differences in EXOC6A dynamics between Supplementary Figure 2 and Figure 3D most likely reflect cell-to-cell variability in dynamic behavior, which is common in live-cell imaging. Both figures were derived from the same stable cell line co-expressing EXOC6A and Myo-Va-GTD. Moreover, our analysis shows that Myo-Va-GTD overexpression does not suppress EXOC6A dynamics, nor is it required for membrane remodeling per se. However, Myo-Va is essential for EXOC6A recruitment to the ciliary vesicle, as shown by the loss of EXOC6A localization in Myo-Va KO cells (Fig. 4A).

(4) The authors show that loss of EXOC6A affects the localization of some transition zone proteins. Does this subsequently lead to defects in transition zone function?

We agree with the reviewer that structural defects in the transition zone (TZ) should be linked to its function. To address this, we examined the localization of two wellcharacterized ciliary membrane-associated proteins: BBS9 and GPR161. Both proteins failed to localize to the cilia in EXOC6A knockout cells, despite proper recruitment in wildtype controls (Fig. 7C). Although we did not examine the exact functions of GPR161 and BBS9, our results suggest that the loss of EXOC6A may impair TZ function, particularly its gating capacity for membrane protein trafficking.

(5) Additional information about how the MKS proteins are regulated by EXOC6A would be helpful to understand the mechanisms by which EXOC6A builds the transition zone. Does EXOC6A directly bind to MKS proteins, or are the MKS proteins delivered by EXOC6A-containing vesicles during ciliogenesis?

We appreciate the reviewers' questions regarding the mechanistic relationship between EXOC6A and MKS module proteins. In this study, we did not explore the mechanism by which EXOC6A constructs the transition zone. This is an interesting topic worthy of further investigation in the future.

Reviewer #3 (Recommendations for the authors):

Recommended modifications:

(1) The co-immunoprecipitation experiments suggest an interaction between EXOC6A and Myo-Va; however, the presence of a faint band in the IgG control raises some uncertainty. To reinforce this conclusion, the authors could demonstrate that the interaction is absent in the EXOC6A knockout cell line.

We thank the reviewer for this careful observation. We acknowledge the presence of a faint Myo-Va signal in the IgG control lane. Myosin-Va is a highly abundant cytoskeletal motor protein and can occasionally exhibit low-level nonspecific binding to agarose beads during immunoprecipitation assays. Importantly, the Myo-Va signal co-immunoprecipitated with endogenous EXOC6A is substantially stronger and specifically enriched compared with the IgG control, supporting a specific interaction.

(2) Figure S5: The partial rescue of the EXOC6A phenotype is not entirely convincing. A statistical test to assess the significance of the observed differences may help to strengthen the authors' conclusion.

We appreciate the reviewer's suggestion to validate the rescue experiment. We have now performed a pairwise two-tailed Student's t-test comparing ciliogenesis efficiency between EXOC6A knockout cells and rescue cells expressing GFP-EXOC6A. As shown in the revised Figure S6 (original Figure S5), re-expression of EXOC6A resulted in a statistically significant recovery of ciliogenesis ($p = 0.0041$). While the rescue is partial—likely due to inherent limitations of plasmid-based expression systems, including variable transfection efficiency and imperfect restoration of endogenous protein levels—the statistically significant improvement confirms that the ciliogenesis defect is specifically caused by EXOC6A loss. Figure S6 and its legend have been updated accordingly.

(3) A detailed description of the EXOC6A knockout strategy should be included.

The Method section has been expanded to include a comprehensive description of the CRISPR/Cas9 - mediated EXOC6A knockout strategy, including sgRNA sequences, genomic target sites, and validation approaches. Additionally, we now include Figure S3,

demonstrating complete loss of EXOC6A protein expression in two independent knockout clones, confirming the efficiency and specificity of the gene-editing strategy.

(4) The labeling in Figure 6 is confusing; assigning a separate letter to each panel would improve clarity.

Figure 6 has been reorganized for clarity: the original panels have been subdivided and relabeled as 6A/6A' and 6B/6B', respectively. The figure legend and all corresponding references in the main text have been updated accordingly.

(5) Lines 109-112: The cell line used is not well described. While experts might understand that Dox is used to induce expression of the transgenes, this should be better explained for non-expert readers.

We have revised the text to clearly explain that doxycycline (Dox) is used to induce transgene expression via a Tet-On inducible system. This clarification has been added to the main text.

(6) Line 180: replace "labels" with "structures".

We have revised the text as suggested.

(7) Line 189: the EXOC6A recruitment to the membrane structures seems to be occurring on a short timescale that should be specified. In this context, "immediately" appears unscientific.

We have revised the sentence to specify that EXOC6A recruitment occurs within seconds, based on our live-cell imaging data, providing a more accurate temporal description.

(8) Lines 280-282: We recommend rewording to soften this statement. Actin and microtubule inhibitors affect the entire cytoskeletal network; more specific experiments would be required to assess whether the transport of vesicles is defective.

We have reworded the statement to indicate that the accumulation of these vesicles at the mother centrioles is highly sensitive to disruption of dynein or microtubules, suggesting that efficient transport of these vesicles may depend on the integrity of the microtubule network. However, more experiments are required to confirm this conclusion.

(9) Lines: 428-433: Similarly, we recommend rewording this statement as it presents the authors' current model, which is in line with the presented data but would require more rigorous investigation.

We have revised this section to describe the mechanism as a working model supported by our data, while acknowledging that further investigation will be required to fully establish the proposed hierarchy and molecular details.

Questions and comments to consider:

(1) 15-30% of cells can form cilia-like structures in the EXOC6A KO cells, although membrane transport should be reduced. It would be interesting to investigate whether these cilia are only formed intracellularly and fail to reach the cell surface.

We thank the reviewer for this insightful question. Using both immunofluorescence and electron microscopy, we observed that a subset of ciliary membranes in EXOC6A KO cells do appear to fuse with the plasma membrane. However, due to the low frequency and heterogeneous morphology of these structures, we were unable to reliably quantify this population.

(2) In the Western blot shown in Figure 4, EXOC6A appears at multiple molecular weights when detected with the anti-EXOC6A antibody. Providing a possible explanation for this

shift would be helpful.

We clarify that the apparent molecular weight shift likely results from gel distortion during electrophoretic separation. Importantly, the specificity of the major EXOC6A band was rigorously validated by its complete absence in EXOC6A knockout lysates, confirming that the detected signal corresponds to EXOC6A.

(3) The Western blot in Figure 5B is not fully convincing; including additional independent blots would be nice.

We thank the reviewer for this suggestion. Figure 5B has been replaced with a blot from an independent experiment, improving clarity and reproducibility.

(4) According to the materials and methods section, siRNA-mediated knockdown of targets was performed using a single siRNA per gene, which could result in off-target effects. It would be advised to use several different siRNAs for a single target to exclude off-target effects, cite references or, in case this has been done.

We appreciate this concern. The siRNAs used in this study were previously validated in our earlier work (Wu et al., Nat Cell Biol 2018), where both specificity and efficiency were rigorously tested. We have now explicitly cited this reference in the Materials and Methods section to justify the selection of these reagents.

(5) The abbreviation CFLEM is uncommon for correlative (fluorescence) light and electron microscopy; the authors should consider using the standard abbreviation CLEM.

We have replaced “CFLEM” with the standard term CLEM (Correlative Light and Electron Microscopy) throughout the manuscript and figure legends.

(6) The term "M-centriole" is uncommon and should at least be introduced. The use of the term "mother centriole" is recommended.

We have replaced “M-centriole” with the standard term “mother centriole” throughout the manuscript and figures.

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